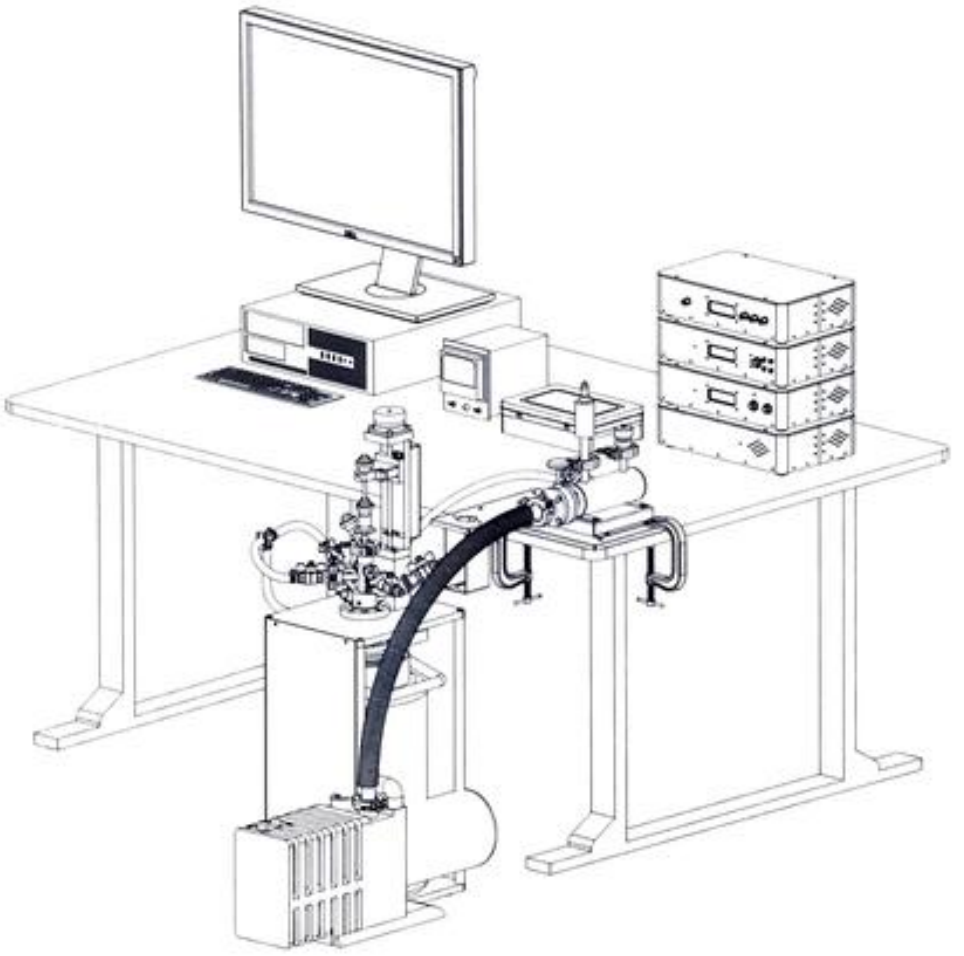
XPLORE 1.2™

Physical Quantities Measurement System (PQMS)

User Manual



Precision-Quazar Tech Pvt. Ltd.
New Delhi, India

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Temperature Controller Module

1.1 Overview

Temperature controller module consists of a micro-processor based electronic control unit, and its driver and command utility. It has two temperature sense channels and a power output channel to drive a 35 Watt heater. The system is calibrated for standard platinum resistance temperature sensor, Pt100. The front and rear view of the electronic unit are shown in Figure 1.1. A screen shot of the associated command utility is shown in Figure 1.2.

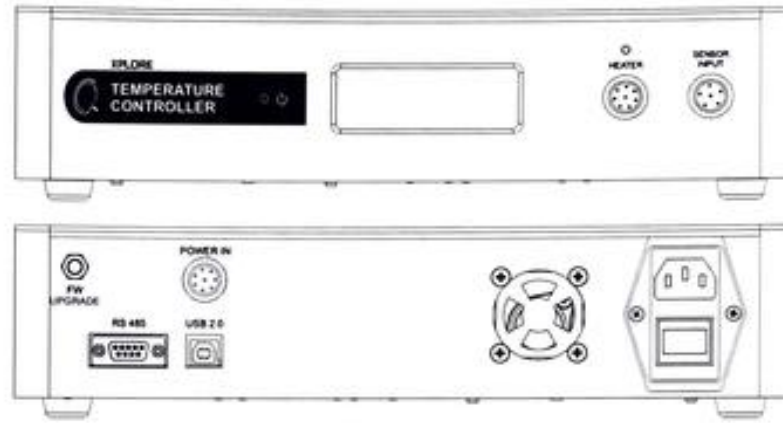


Figure 1.1: The front and rear view of the micro-processor based electronic unit of the temperature controller module.

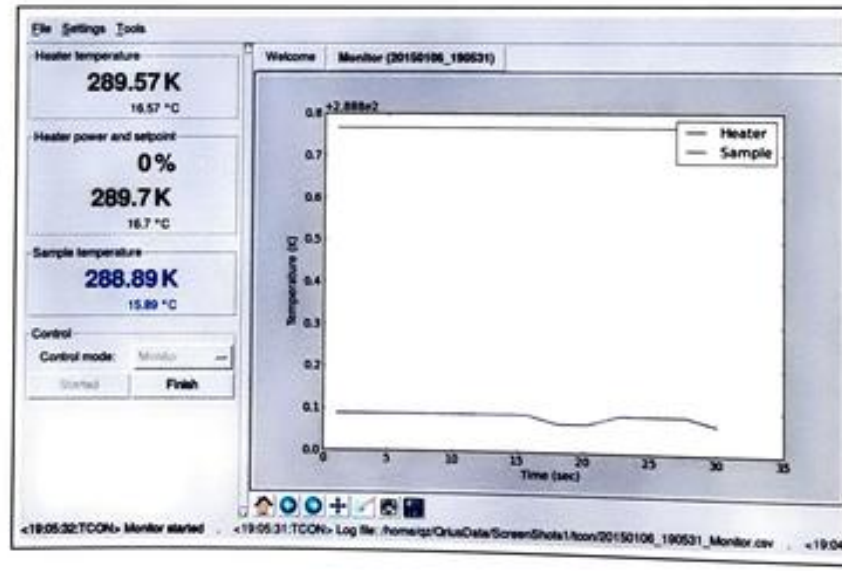


Figure 1.2: A screen shot of the command utility for temperature controller module. Various control parameters can be accessed by opening their respective dialog boxes from the **Settings** menu in the menu bar.

The temperature control algorithm is implemented using an intelligent PID loop, which senses heater temperature and accordingly sets the output power to reach a target temperature. The current values of the heater temperature and sample temperature are displayed in the left panel of the utility window, as well as in the LCD of the electronic unit. Additionally current set-point and heater power are also shown on the computer screen. The temperature controller can run in four different modes, viz. monitor, isothermal, linear ramp, and stepped ramp. They are explained in details in the following sections.

1.2 Isothermal mode

The temperature controller is capable of maintaining the heater temperature at a fixed value for an indefinite time. To set the target temperature, open the isothermal temperature settings dialog box by clicking **Settings → Isothermal settings** in the menu bar of the temperature controller utility window (Figure 1.2). The opened dialog is shown in Figure 1.3.



Figure 1.3: Dialog box to set the temperature set-point of an isothermal run

The working temperature range is 80 K to 450 K. To apply the new temperature set-point, click **File → Apply**. To ignore, choose **File → Cancel**.

i The isothermal control has to be explicitly activated by selecting **Control mode → Isothermal** in temperature controller utility window (Figure 1.2) and clicking the **Start** button. In case of any communication error, a descriptive message is displayed in the status bar at the bottom of the window. Click **Finish** to end an isothermal session.

i If the module is being used by another higher-level module; e.g. resistivity vs. temperature, susceptibility vs. temperature; it is not possible to terminate an ongoing run using the **Finish** button. A message **"Cannot stop! Running in slave mode."** would appear in the status bar.

1.3 Linear ramp mode

The temperature controller is capable of ramping heater temperature at a fixed rate till it reaches a user defined final temperature. To configure the final temperature and ramp rate, open the linear ramp settings dialog box by clicking **Settings → Linear ramp settings** in the menu bar of the temperature controller utility window (Figure 1.2). The opened dialog is shown in Figure 1.4.

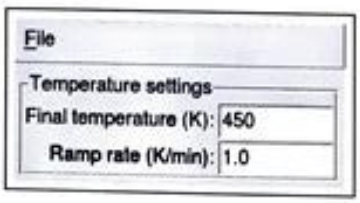


Figure 1.4: Dialog box to set final temperature and ramp rate of a linear temperature ramp

The maximum value of the final temperature is 450 K. The ramp-rate can be set from 0.1 K/min to 10 K/min in steps of 0.1 K/min. To apply the ramp settings, click **File → Apply**. To ignore, choose **File → Cancel**.

i The linear temperature ramp has to be explicitly activated by selecting

Control mode → Linear ramp in temperature controller utility window (Figure 1.2) and clicking the **Start** button. **The current heater temperature is taken to be the starting point of the linear ramp.** In case of any communication error, a descriptive message is displayed in the status bar at the bottom of the window. Click **Finish** to end a run.

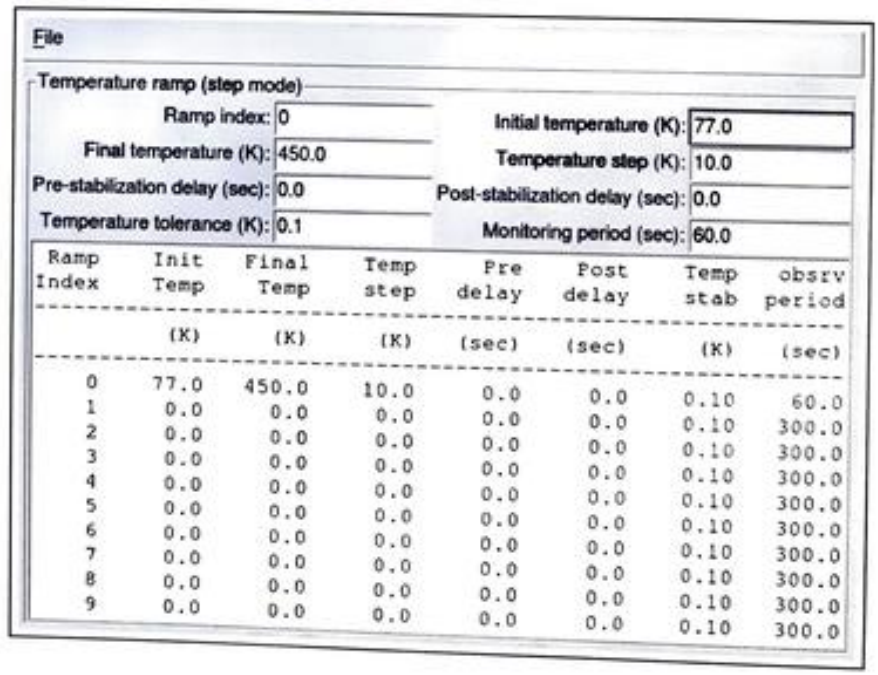
i It is possible to achieve a negative ramp by entering a negative value in the ramp rate. However, the temperature controller does not have any active cooling mechanism. Thus the maximum slope is limited by the natural cooling rate.

i If the module is being used by another higher-level module; e.g. resistivity vs. temperature, susceptibility vs. temperature; it is not possible to terminate an ongoing run using the **Finish** button. A message **"Cannot stop! Running in slave mode."** would appear in the status bar.

1.4 Stepped ramp mode

It is sometimes required to stabilize the temperature at a given value, do some measurement, and proceed to the next temperature. This is known as stepped ramping.

The temperature controller module is fully capable of running in stepped ramp mode with configurable stability criteria. To open the stepped ramp configuration dialog, click **Settings → Stepped ramp settings** in the menu bar of the temperature controller utility window (Figure 1.2). The opened dialog is shown in Figure 1.5.



Ramp Index	Init Temp (K)	Final Temp (K)	Temp step (K)	Pre delay (sec)	Post delay (sec)	Temp stab (K)	obsrv period (sec)
0	77.0	450.0	10.0	0.0	0.0	0.10	60.0
1	0.0	0.0	0.0	0.0	0.0	0.10	300.0
2	0.0	0.0	0.0	0.0	0.0	0.10	300.0
3	0.0	0.0	0.0	0.0	0.0	0.10	300.0
4	0.0	0.0	0.0	0.0	0.0	0.10	300.0
5	0.0	0.0	0.0	0.0	0.0	0.10	300.0
6	0.0	0.0	0.0	0.0	0.0	0.10	300.0
7	0.0	0.0	0.0	0.0	0.0	0.10	300.0
8	0.0	0.0	0.0	0.0	0.0	0.10	300.0
9	0.0	0.0	0.0	0.0	0.0	0.10	300.0

Figure 1.5: Dialog to set various parameters of a stepped ramp, e.g. temperature range, step size, and stability criteria.

- **Initial temperature** and **final temperature** signify the heater

temperature range over which step size and stability criteria are defined.

- **Temperature step** sets the jump in heater temperature between two successive stable plateau.
- **Temperature tolerance** and **Monitoring period** specify stability criteria. The temperature is said to be stable if the sample temperature does not vary more than the specified tolerance over the monitoring period.
- **Pre-stabilization delay** and **Post-stabilization delay** are two arbitrary delays that can be inserted before and after the execution of temperature stabilization loop.
- **Ramp index** selects a ramp, i.e. a temperature range, for which the above parameters are entered. It is possible to have different step sizes, pre- and post-stabilization delays, temperature tolerances, and monitoring periods in different temperature ranges.

i **Initial temperature**, **Final temperature**, and **Temperature step** are specified in terms of **heater temperature**, while **Temperature tolerance** is specified in terms of **Sample temperature**.

i The stepped temperature ramp has to be explicitly activated by selecting **Control mode → Step ramp** in temperature controller utility window (Figure 1.2) and clicking the **Start** button. In case of any communication error, a descriptive message is displayed in the status bar at the bottom of the window. Click **Finish** to end a run.

i During a run, **Ramp index**, **Heater setpoint**, and **State** are shown in the left panel. A temperature step chronologically goes through **Pre delay**, **Stabilizing**, **Post delay**, and **Stable** states before continuing to the next set-point. During stabilization, sample temperature fluctuation is also displayed for easy reference.

i If the module is being used by another higher-level module; e.g. resistivity vs. temperature, susceptibility vs. temperature; it is not possible to terminate an ongoing run using the **Finish** button. A message **"Cannot stop! Running in slave mode."** would appear in the status bar.

1.5 Monitor mode

The temperature controller can also be used as a simple temperature recorder without any active control loop. Both sample and heater temperature are plotted on the computer screen as a function of time. This is useful while cooling down the system.

i Monitoring has to be explicitly started by selecting **Control mode → Monitor** in temperature controller utility window (Figure 1.2) and clicking the **Start** button. In case of any communication error, a de-

scriptive message is displayed in the status bar at the bottom of the window. Click **Finish** to end.

1.6 PID parameters

Wrong settings can make the temperature controller to oscillate.

The temperature controller uses an intelligent PID algorithm to set heater power based on the current set-point and observed heater temperature. The heater power H, in terms of % of maximum power, is calculated as

$$H = 100 \times \left(Pe + I \int e dt + D \frac{de}{dt} \right) \tag{1.1}$$

where e is the difference between actual heater temperature and temperature set-point, and P, I, and D are control parameters. To modify these settings, open the temperature control parameters dialog box by clicking **Settings → PID settings** in the menu bar of the temperature controller utility window (Figure 1.2). The opened dialog is shown in Figure 1.6.

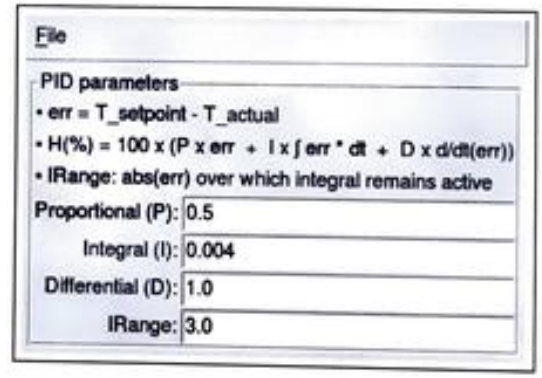


Figure 1.6: Dialog to set PID control parameters

Factory configured values are pre-filled in the dialog. To apply the new parameters, click **File → Apply**. To ignore, choose **File → Cancel**.

The integral control is active only when the difference between the actual temperature and set-point is within the specified **IRange**.

In case of any communication error, a descriptive message is displayed in the status bar at the bottom of the window.

1.7 Data files

As soon as a run is initiated, a log file is opened in the temperature controller's default data folder, **/home/pqms/QriusData/Sample1/tcon**. The name of the file is derived from current date **YYYYMMDD** and time **HHMMSS** as **YYYYMMDD_HHMMSS_Mode.csv**, where **Mode** can be one of the following – **Monitor**, **Isothermal**, **LinearRamp**, and **StepRamp**.

i A few lines at the beginning of the file contains information about sample details and column headings. These lines start with a #. Thereafter each row contains time stamp, heater temperature, sample temperature, heater set-point, and heater power separated by comma.

i The default destination path of the recorded data can be changed by an user using the following guidelines.

- **/home/pqms** is the home folder corresponding to **pqms** login id. If some other login id is used, the path changes accordingly.
- **QriusData** is the root of the default destination folder. This can be changed from **Settings → Global settings** in the menu bar of Qrius master window.
- **Sample1** is the default sample folder generated by concatenation of default sample name (**Sample**) and sample id (**1**). This can be changed from **Sample settings** dialog from the Qrius master window.
- **tcon** represents a folder dedicated for temperature controller. **It cannot be changed.**

i The recorded file, being in **csv** format, can be directly opened in OpenOffice, Excel, Origin, Matlab, and other popular analysis tools. At the end of a run, another file with **xpl** extension is created, which can be opened by Qrius Analyzer.

1.8 Save and recall settings

It is always useful to save instrument settings for future reference. This is done through method files. A method file is identified with its **xmt** extension. To create a method file, first set all parameters using different dialog boxes described in previous sections, and click **File → Save method** in the menu bar of the temperature controller utility window (Figure 1.2). Enter the file name in the opened dialog. To load a stored method, use **File → Open method**.

i By default the method file goes to **/home/pqms/QriusData/Sample1/method** folder.
This can be changed in a similar way described for the data files.

i If the method file is not for temperature controller, an error message "**Method failed. Not a TCON method**" is shown in the status bar.

1.9 Technical specification

Parameter	Value
Measurement	
Temperature range	77 K – 450 K
Sensor type	Pt100
Temperature resolution	0.01 K
Control	
Temperature stability	Better than ±0.1 K in isothermal mode
Temperature ramp-rate	0.1 K/min – 10 K/min in steps of 0.1 K/min
Control algorithm	intelligent PID
PID parameters	User configurable
Heater	
Heater power	35 Watt
Others	
Computer connectivity	USB

I-V Source & Measurement Module

2.1 Overview

I-V source & measurement module consists of a micro-processor controlled electronic unit, and its associated driver and utility software. The electronics has one output channel which can be configured to supply either a constant DC voltage or a constant DC current. It also has three input channels to measure supplied current, driving voltage, and an external voltage. Hence it is also referred as DC source-meter. The front and rear view of the electronic unit and its associated control utility are respectively shown in Figure 2.1 and 2.2.

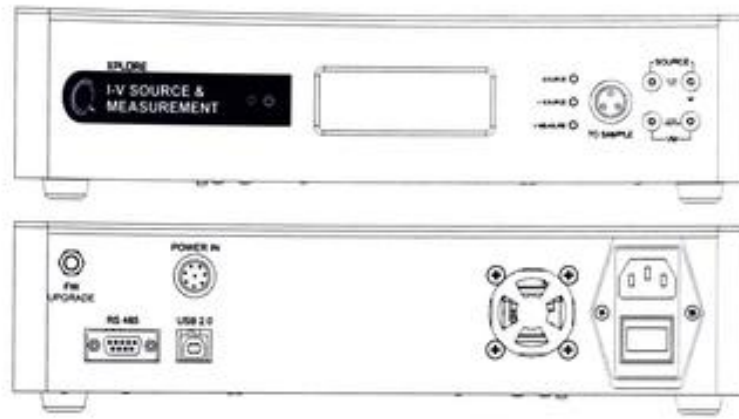


Figure 2.1: Source-meter front & rear View

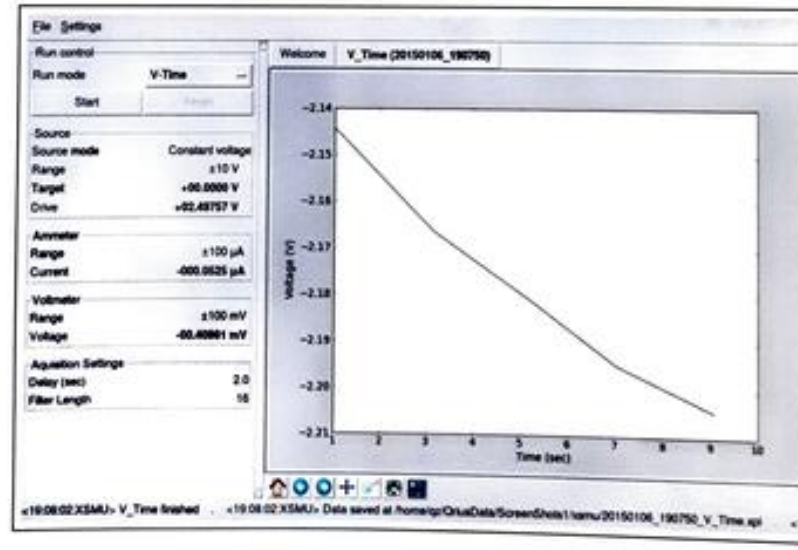


Figure 2.2

This module may be used to study current vs. voltage characteristic of various materials and devices using two- and four-probe connection topology. It can also be used to record current, voltage, and resistance as a function of time. It is worth mentioning that the module can intelligently switch between constant current and constant voltage source mode depending on the attached load making it possible to characterize non-linear devices. The following sections explain how to configure various options to perform these operations.

2.2 Resistance measurement settings

Source and measurement module is comfortably capable of measuring resistance in the range of 1 mΩ to 100 MΩ. This wide range is achieved by switching between current and voltage excitation. For low resistance measurement (up to 10 kΩ), current source with four-probe connection topology is preferred, while for high resistances (above 10 kΩ) voltage source with two-probe connection topology is used.

In constant current mode, a known current is sent through the test sample, and resulting voltage drop is measured, while in constant voltage mode, the test sample is subjected to a constant driving voltage, and resulting current is measured. The electronic unit can intelligently choose between these two modes based on the attached load.

To specify maximum limits of applied voltage, current, and power, open resistance measurement settings dialog box by clicking **Settings → Resistance measurement settings** on the menu bar of the utility window (Figure 2.2). The opened dialog is shown in Figure 2.3.

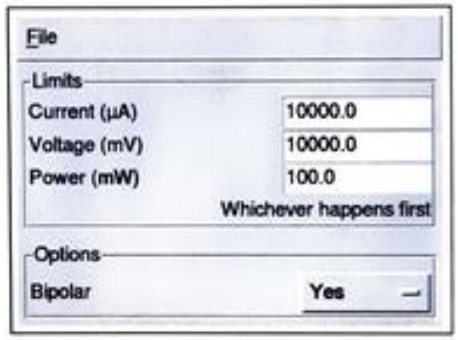


Figure 2.3: Dialog box to configure voltage, current, and power limit during resistance measurement

The description of various input parameters are as follows.

- **Current**, **Voltage**, and **Power** specify the maximum limits of these quantities that would be applied on the sample. Only one of these limits is satisfied at one time. The source accordingly switches to constant current or voltage mode to achieve required regulation.
- If **Bipolar** option is set to **Yes**, then the resistance is measured using both forward and reverse current, and by taking their average. This ensures that offsets present in voltage and current measurements do not affect the final measurement of resistance. However, if the device cannot handle reverse polarity, the current reversal can be turned off by setting **Bipolar** to **No**.

Finally to confirm the settings, click **File → Apply** in the menu bar. To discard, click **File → Cancel**.

i Resistance measurement has to be explicitly activated by selecting **Run mode → R-Time** in source and measurement utility window (Figure 2.2) and clicking the **Start** button. This starts plotting of resistance

as a function of time. In case of any communication error, a descriptive message is displayed in the status bar at the bottom of the window. To terminate the run, click **Finish**.



Power is calculated by multiplying drive voltage to resulting current. If this hits power limit, a binary search is performed on the drive voltage/current to keep the power within range. Generally a few seconds is required to perform power regulation.

2.3 I-V measurement settings

I-V measurement is used to characterize current vs. voltage relationship of a test sample. It starts from zero, and proceeds towards the positive direction. After completing positive ramp, it restarts from zero in the negative direction. To configure the extent of the sweep and step sizes, open I-V measurement settings dialog by clicking **Settings → I-V measurement settings** on the menu bar of the utility window (Figure 2.2). The opened dialog is shown in Figure 2.4.

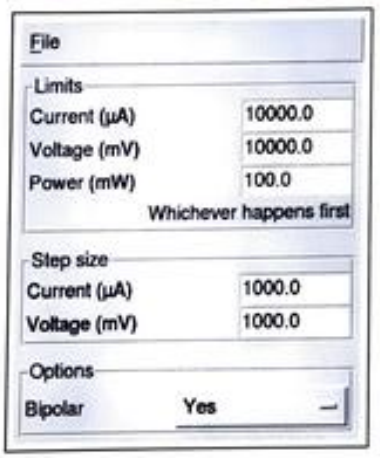


Figure 2.4: Dialog box to configure the limits and step sizes for an I-V sweep

- **Current**, **Voltage**, and **Power** entries in **Limits** section specify the extent of I-V sweep. They are applicable to both positive and negative directions. If any of these limits is reached, the I-V sweep in that direction is ended.
- **Current** and **Voltage** input boxes in **Step size** section specify the maximum possible increment in drive current and voltage across the source terminals. The module automatically switches between constant current and voltage sources to achieve this.
- If **Bipolar** option is set to **Yes** then the IV sweep will be performed in both positive and negative directions. It the test sample cannot withstand reverse polarity, set this to **No**.

Finally to confirm the settings, click **File → Apply** in the menu bar. To discard, click **File → Cancel**.



Voltage limit and step size refers to the drive voltage at the source ter-

minals of the electronic unit. However, I-V plot refers to the voltage recorded through the measurement terminals.

2.4 Acquisition settings

Acquisition settings dialog box can be accessed by clicking **Settings → Acquisition settings** on the menu bar of the utility window (Figure 2.2). The opened dialog is shown in Figure 2.5.

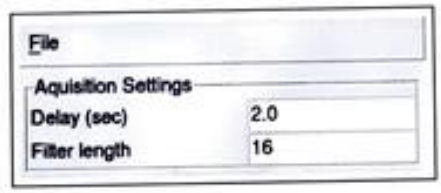


Figure 2.5: Dialog box to configure delay time and filter length parameters for every acquisition

- **Delay (sec)** is the time in seconds for which the system waits after applying any current or voltage source before taking any kind of measurement. Default value is 2 seconds.
- **Filter length** is the number of readings in either current or voltage measurement over which averaging is performed. The system, thereafter, uses this averaged value. Default value is 16.

Finally to confirm the settings, click **File → Apply** in the menu bar. To discard, click **File → Cancel**.

2.5 Source parameters

The output channel of the electronic unit can either be configured as a constant voltage source, or as a constant current source. To set the source mode, open the source parameters dialog by clicking **Settings → Source parameters** on the menu bar of the utility window (Figure 2.2). The opened dialog is shown in Figure 2.6.

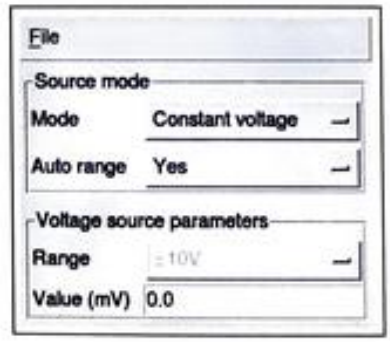


Figure 2.6: Dialog box to use constant current or constant voltage source

Various input parameters are described below.

- **Mode** signifies source configuration which can be set to **Constant current** or **Constant voltage**.

- **Auto range** enables automatic selection of a suitable source range depending on the entered **Value**. This option is set to **Yes** by default. To turn it off select **No**.
- If **Auto range** is set to **No**, the source range can be manually selected from the drop-down **Range** menu. Possible ranges are **±100 µA**, **±1 mA**, and **±10 mA** for the current source, and **±10 V** for the voltage source. If **Auto range** is set to **Yes** this option is deactivated.
- **Value** signifies the set-point of the source in specified unit. For current source, it is in μA, while for voltage source it is in mV.

Finally to confirm the settings, click **File → Apply** in the menu bar. To discard, click **File → Cancel**.



These settings will be overridden during I-V characterization and resistance measurement by those specified in **I-V measurement settings**, and **Resistance measurement settings** dialog box, respectively.

2.6 Meter settings

Meter settings dialog box can be accessed by clicking **Settings → Meter settings** on the menu bar of the utility window (Figure 2.2). The opened dialog is shown in Figure 2.7.

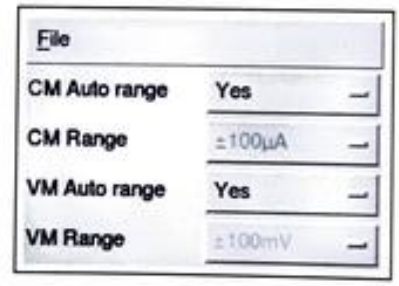


Figure 2.7: Dialog box to configure the current and voltage measurement unit

- **CM Auto range** enables automatic selection of a suitable current measurement range depending upon the current being drawn. This option is set to **Yes** by default. To turn it off select **No**.



If the source is set to constant current mode then this option and the next is deactivated.

- If **CM Auto range** is set to **No**, the current measurement range can be manually selected from the drop-down **CM Range** menu. Possible ranges are **±100 µA**, **±1 mA** and **±10 mA**. If **CM Auto range** is set to **Yes** this option is deactivated.
- **VM Auto range** enables automatic selection of a suitable voltage measurement range depending upon the voltage at the measurement terminals. This option is set to **Yes** by default. To turn it off select **No**.

- If **VM Auto range** is set to **No**, the voltage measurement range can be manually selected from the drop-down **VM Range** menu. Possible ranges are **±100 mV**, **±1 V**, and **±1 V**. If **VM Auto range** is set to **Yes** this option is deactivated.

Finally to confirm the settings, click **File → Apply** in the menu bar. To discard, click **File → Cancel**.

2.7 Intelligent switching between constant current and voltage source

The system first opts for constant current mode and tries to supply current specified in the **Current** limit to the sample. If the voltage supplied by the source to drive such current is less than the voltage specified in the **Voltage** limit then the source mode is not changed. However, if the voltage supplied by the source is greater than the voltage specified in the **Voltage** limit, then the source mode is changed to constant voltage and the source supplies the voltage specified in the **Voltage** limit across the source terminals.

If at any point the power ($I \times V$) exceeds the power specified in the **Power** limit then the current/voltage excitation limits are reduced suitably to comply with the **Power** limit. Thus the algorithm ensures that the current/voltage excitation is within the zone illustrated in Figure 2.8.

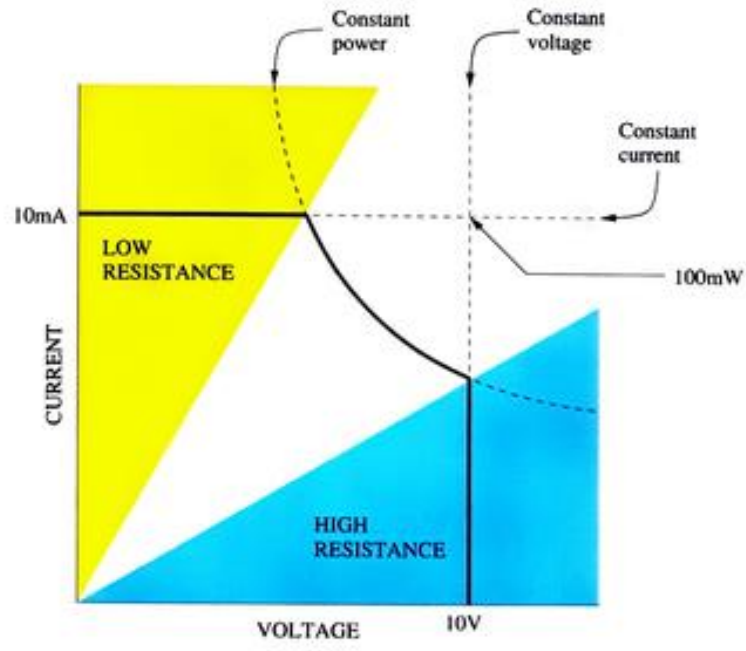


Figure 2.8

2.8 Data files

As soon as a run is initiated, a log file is opened in the SMU's default data folder, **/home/pqms/QriusData/Sample1/xsmu**. The name of

the file is derived from current date **YYYYMMDD** and time **HHMMSS** as **YYYYMMDD_HHMMSS_Mode.csv**, where **Mode** can be one of the following – **I_ Time**, **V_ Time**, **IV**, and **R_ Time**.

i A few lines at the beginning of the file contains information about sample details and column headings. These lines start with a #. Thereafter each row contains time stamp, current, measurement voltage, source voltage, and resistance separated by comma.

i The default destination path of the recorded data can be changed by an user using the following guidelines.

- **/home/pqms** is the home folder corresponding to **pqms** login id. If some other login id is used, the path changes accordingly.
- **QriusData** is the root of the default destination folder. This can be changed from **Settings → Global settings** in the menu bar of Qrius master window.
- **Sample1** is the default sample folder generated by concatenation of default sample name (**Sample**) and sample id (**1**). This can be changed from **Sample settings** dialog from the Qrius master window.
- **xsmu** represents a folder dedicated for source and measurement unit. **It cannot be changed.**

i The recorded file, being in **csv** format, can be directly opened in OpenOffice, Excel, Origin, Matlab, and other popular analysis tools. At the end of a run, another file with **xpl** extension is created, which can be opened by Qrius Analyzer.

2.9 Save and recall settings

It is always useful to save instrument settings for future reference. This is done through method files. A method file is identified with its **xmt** extension. To create a method file, first set all parameters using different dialog boxes described in previous sections, and click **File → Save method** in the menu bar of the XSMU utility window (Figure 2.2). Enter the file name in the opened dialog. To load a stored method, use **File → Open method**.

i By default the method file goes to **/home/pqms/QriusData/Sample1/method** folder.
This can be changed in a similar way described for the data files.

i If the method file is not for XSMU, an error message "**Method failed. Not a XSMU method**" is shown in the status bar.

2.10 XSMU Specifications

Parameter	Value
Source mode	Constant current or voltage
Current source specification	
Current source range(s)	100 μA, 1 mA, 10 mA
Current set-point resolution	Better than 0.05 % of full-scale
Voltage compliance	10V
Voltage source specification	
Voltage source range(s)	10 V
Voltage set-point resolution	Better than 0.005 % of full-scale
Current compliance	10 mA
Ammeter specification	
Current measurement range(s)	100 μA, 1 mA, 10 mA
Current measurement resolution	6½ digit
Voltmeter specification	
Voltage measurement range(s)	100 mV, 1 V, 10 V
Voltage measurement resolution	6½ digit
Input impedance	$> 10^{12}\,\Omega$
Ohmmeter specification	
Resistance measurement range(s)	10 Ω, 100 Ω, 1 kΩ, 10 kΩ, 100 kΩ, 1 MΩ, 10 MΩ
Resistance measurement resolution	6½ digit
Others	
Computer connectivity	USB

Lock-in Amplifier Module

3.1 Overview

The Xplore Lock-in Amplifier (XLIA) module consists of a in-built reference generator and a two channel lock-in amplifier, both contained in a micro processor based electronic unit. The reference generator of XLIA is capable of supplying AC signal with frequency ranging from 10 Hz to 10 kHz. The amplitude of the signal can be either a fixed voltage or a fixed current. The lock-in amplifier can lock the measurement signal with the reference signal and measure the in-phase component and quadrature component simultaneously. The front and back view of the electronic control unit is shown in Figure 3.1. The screen shot of the associated command utility is shown in Figure 3.2.

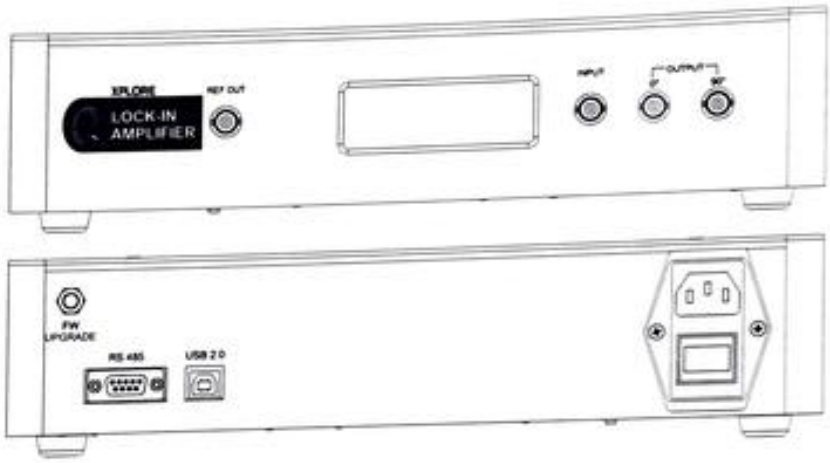


Figure 3.1: The front and back view of the electronic unit of the XLIA.

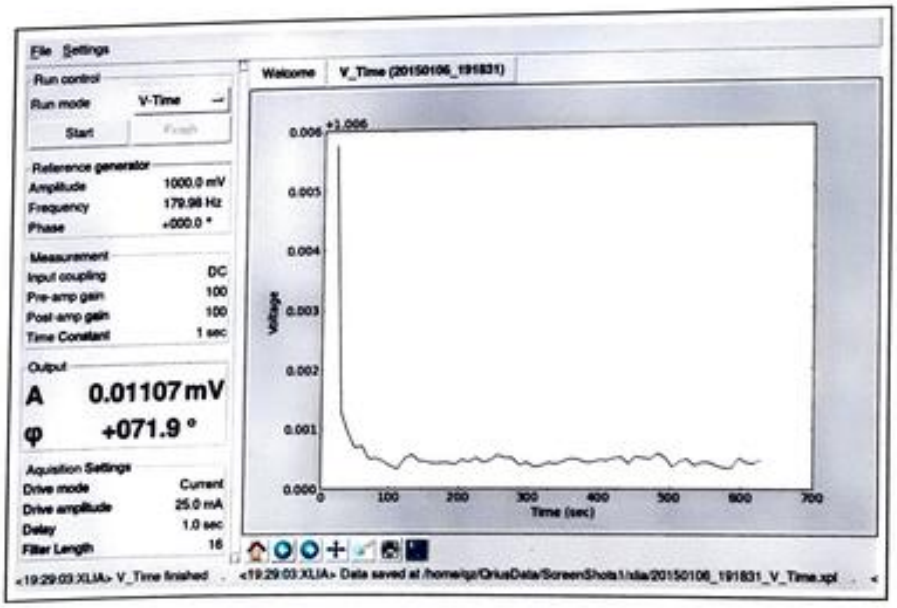


Figure 3.2: A screen shot of the command utility for the lock-in amplifier. Various parameters for reference and measurement settings can be accessed by opening their respective settings dialog box from the **Settings** menu on the menu bar.

3.2 Theory of operation

A lock-in amplifier is a hardware implementation of Fourier transformation. It measures amplitude and phase of a signal at a given frequency exploiting orthogonal nature of sin and cos functions. It has a reference generator, multiplier, integrator, and pre- and post-amplification stages to perform its operation. A block-diagram of a lock-in amplifier is shown in Figure 3.3.

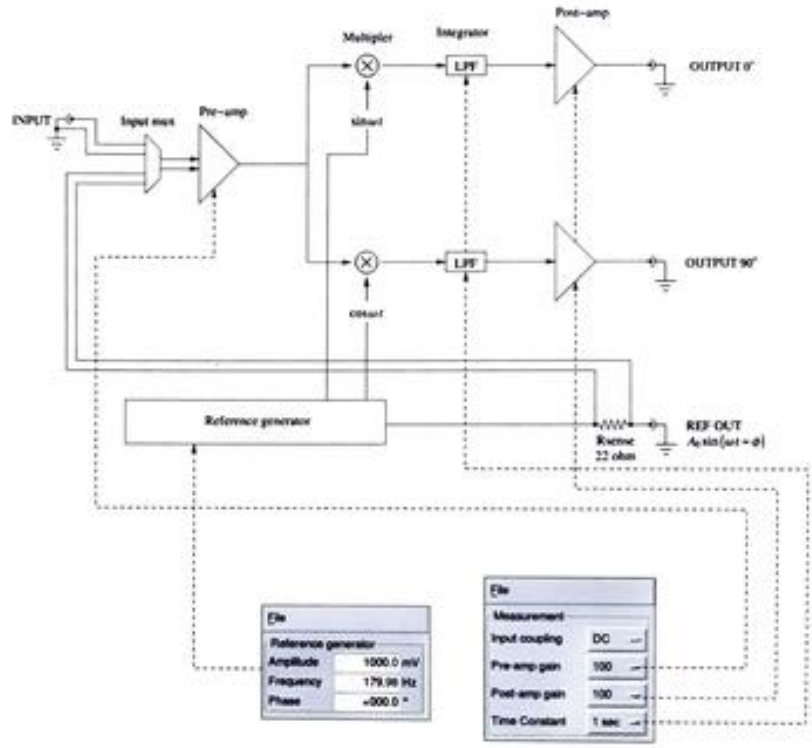


Figure 3.3: Block diagram of a lock-in amplifier showing its reference generator and demodulator circuits.

Reference generator It generates three sinusoidal signals locked at a given frequency, $\omega = 2\pi f$, specified by the user.

1. Two of these signals, viz. $\sin\omega t$ and $\cos\omega t$, are internally fed to in-phase (0°) and quadrature (90°) multiplier circuits.
2. The third signal, $A_0\sin(\omega t + \phi)$, is available outside the box through **REF OUT** terminal on the front panel. Its amplitude A_0 and relative phase ϕ with respect to internal $\sin\omega t$ can be set from **Reference settings** dialog box as described in Section 3.3.

Demodulator It refers to the processing circuitry of the input signal, fed through the **INPUT** terminal on the front panel of the lock-in amplifier. The input first goes through a pre-amplification stage which offers a gain of **1**, **10**, or **100**. The amplified signal, which is mostly AC, is then bifurcated and fed two parallel processing paths, each consists of a multiplier, an integrator, and a post-amplification stage.

In this path, the signal is first multiplied with an internal reference signal (sin for in-phase, and cos for quadrature channel) before being fed

to a second order low-pass filter for integration. The filter has three preset time constants, viz. **2 ms**, **5 ms**, and **1 sec**. The post-amplifier further amplifies the DC output of the integrator with a gain of **1**, **10**, or **100**.

Both in-phase and quadrature demodulated outputs are then simultaneously measured by an **ADC**. These are also available at 0° and 90° ports at the front panel.



The lock-in amplifier can record reference current by measuring voltage drop across an internal 22Ω current sense resistor connected in series with **REF OUT**. This also limits the output short circuit current to about 45 mA.

3.3 Reference settings

A sinusoidal AC signal is defined by three parameters; amplitude, frequency, and phase (relative to some reference). To set these parameters of the signal coming out from the REF OUT port of the one has to open the reference settings dialog box by clicking **Settings → Reference settings** on the menu bar of the XLIA module window. This will open the dialog box shown in Figure 3.4 .

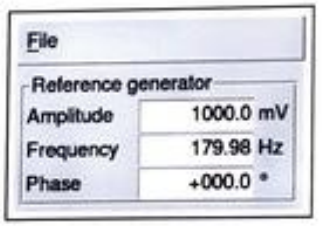


Figure 3.4: Dialog to set amplitude, frequency and phase of the reference signal

- **Amplitude** is the amplitude of the sine wave signal generated across the terminals of the REF OUT port. Its range is 0 - 1000 mV.
- **Frequency** is the frequency of the sine wave signal generated across the terminals of the REF OUT port. Its range in 10 - 10000 Hz.
- **Phase** is the phase difference in degrees of the sine wave signal generated across the terminals of the REF OUT port with respect to the internal reference.

Finally click **File → Apply** in the menu bar of the dialog box to confirm the settings and close the dialog box. Or click **File → Cancel** to discard the changes.



The **Amplitude**, **Frequency**, and **Phase** settings mentioned above get reflected on the terminals of the REF OUT port as soon as the user enters them. However, during a run (such as ...) the **Amplitude** is overridden by the one mentioned in the ... .

3.4 Measurement settings

When the user is measuring a signal manually, he/she can set the pre and post amplification gain and the integration time of the lock-in amplifier (See Section 3.2). Additionally one can choose to apply AC coupling to the signal being measured. These settings can be accessed from the measurement settings dialog box (Figure 3.5). To open it click **Settings → Measurement settings** on the menu bar of the XLIA utility window.

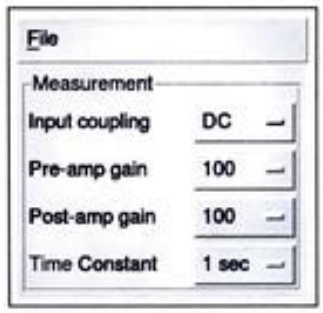


Figure 3.5: Dialog to change the various settings of the measurement channel.

- If **Input coupling** is set to **AC** the signal is passed through a single pole high pass filter with cutoff frequency at 1 Hz. This removes any DC offset present in the input signal. **DC** coupling bypasses this process and unmodified signal is sent for measurement. **AC** coupling should be used only if user has introduced some DC offset in the signal by design.
- **Pre-amp Gain** is the gain of the signal before it is fed to the multiplier channel. Its value can be 1, 10 or 100
- **Post-amp Gain** is the gain of the signal after it comes out of the multiplier channel. Its value can be 1, 10 or 100
- **Time constant** is the duration for which the input signal multiplied with the internal reference is integrated with respect to time. This value should be at least 10 times the time period of the signal. Possible choices are 1 sec, 5 ms, 2 ms.

Finally, click **File → Apply** in the menu bar of the dialog box to confirm the settings and close the dialog box. Or click **File → Cancel** to discard the changes and close the dialog box.

3.5 Frequency sweep (voltage v/s frequency ramp) settings

The system is capable of performing frequency sweep (voltage v/s frequency ramp), which is useful when one has to study the response of material (electric or magnetic) with respect to the frequency of the AC signal applied. In XPLORE 1.2™ system the user has the option to increase the the frequency either uniformly or logarithmically. To open the frequency sweep settings dialog box (Figure 3.6) click **Settings → V-F ramp settings** on the menu bar of the XLIA module window.

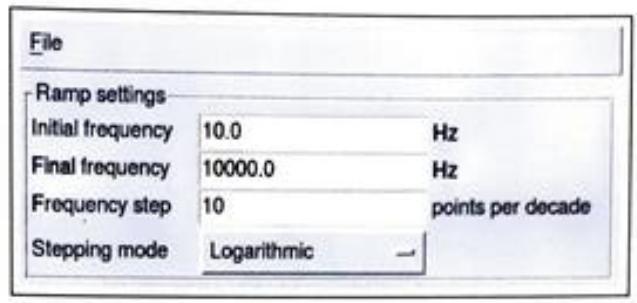


Figure 3.6: Dialog to set the various parameters for the voltage v/s frequency ramp.

- **Initial frequency** is the starting frequency of the voltage v/s frequency ramp.
- **Final frequency** is the final frequency of the voltage v/s frequency ramp.
- **Frequency step** is the size of the frequency intervals of the ramp.
- If **Stepping mode** is selected **Linear**, then the frequency is increased uniformly by the value mentioned in the **Frequency step**.

  If **Stepping mode** is selected **Logarithmic**, then the frequency is increase logarithmically, i.e, the actual frequency step size is increased exponentially. The value mentioned in the **Frequency step** then becomes the number of points in every decade. For example, if **Initial frequency** is 10 Hz,
- **Final frequency** is 10000 Hz, and
- **Frequency step** is 10 per decade, then The program will take 10 readings between 10 Hz and 100 Hz, 10 readings between 100 Hz and 1000 Hz, and 10 readings between 1000 Hz 10000 Hz.

Finally, click **File → Apply** in the menu bar of the dialog box to confirm the settings and close the dialog box. Or click **File → Cancel** to discard the changes and close the dialog box.

3.6 Acquisition settings

To open the acquisition settings dialog box (Figure 3.7) click **Settings → Acquisition settings** on the menu bar of the XLIA module window.

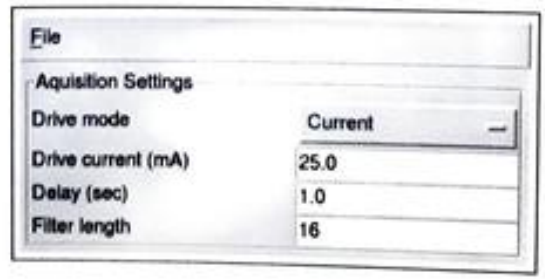


Figure 3.7: Dialog to change the acquisition settings.

- **Drive mode** : If selected **Current**, the system will try to drive an AC signal with a constant amplitude in current through the load connected at the REF OUT port. The amplitude in current will be maintained even if the load resistance changes during the run. If selected **Voltage**, the system will apply an AC signal with a constant amplitude in voltage across the terminals of the REF OUT port. The amplitude in voltage will be maintained even if the load resistance changes during the run.
- **Drive current/voltage** is the value of current/voltage amplitude in mA/mV. In short circuit condition, the maximum current amplitude the device can supply is 40 mA. In open circuit condition the maximum voltage amplitude that the device can apply is 1.0 V.
- **Delay (sec)** is the time in seconds for which the system waits after applying any current or voltage source before taking any kind of measurement. Default value is 2 seconds.
- **Filter length** is the number of readings in either current or voltage measurement over which averaging is performed. The system, thereafter, uses this averaged value. Default value is 16.

Finally, click **File → Apply** in the menu bar of the dialog box to confirm the settings and close the dialog box. Or click **File → Cancel** to discard the changes and close the dialog box.



The above settings are valid only during a run.

3.7 Data files

As soon as a run is initiated, a log file is opened in the XLIA's default data folder, **/home/pqms/QriusData/Sample1/xlia**. The name of the file is derived from current date **YYYYMMDD** and time **HHMMSS** as **YYYYMMDD_HHMMSS_Mode.csv**, where **Mode** can be either **V_Time** or **VF**.

i A few lines at the beginning of the file contains information about sample details and column headings. These lines start with a **#**. Thereafter each row contains time stamp, reference frequency, reference amplitude, reference phase, (drive) current amplitude, (drive) current phase, signal amplitude, and signal phase separated by comma.

i The default destination path of the recorded data can be changed by an user using the following guidelines.

- **/home/pqms** is the home folder corresponding to **pqms** login id. If some other login id is used, the path changes accordingly.
- **QriusData** is the root of the default destination folder. This can be changed from **Settings → Global settings** in the menu bar of Qrius master window.
- **Sample1** is the default sample folder generated by concatenation of default sample name (**Sample**) and sample id (**1**). This can be changed from **Sample settings** dialog from the Qrius master window.
- **xlia** represents a folder dedicated for lock-in amplifier unit. **It cannot be changed.**

i The recorded file, being in **csv** format, can be directly opened in OpenOffice, Excel, Origin, Matlab, and other popular analysis tools. At the end of a run, another file with **xpl** extension is created, which can be opened by Qrius Analyzer.

3.8 Save and recall settings

It is always useful to save instrument settings for future reference. This is done through method files. A method file is identified with its **xmt** extension. To create a method file, first set all parameters using different dialog boxes described in previous sections, and click **File → Save method** in the menu bar of the XSMU utility window (Figure 2.2). Enter the file name in the opened dialog. To load a stored method, use **File → Open method**.

i By default the method file goes to **/home/pqms/QriusData/Sample1/method** folder.
This can be changed in a similar way described for the data files.

i If the method file is not for XLIA, an error message "**Method failed. Not a XLIA method**" is shown in the status bar.

3.9 XLIA specifications

Parameter	Value
Primary driver	
Frequency range(s)	10 Hz – 10 KHz
Reference output	2V peak-to-peak
Reference generation	Direct digital synthesis
Lock-in detection	
Integration time	1 sec
Pre-amp gain(s)	1, 10, 100
Post-amp gain(s)	1, 10, 100
Measurement	
Measurement resolution	5½ digit
Measured quantities	In phase, quadrature phase, amplitude, and phase.
Others	
Computer connectivity	USB

Sample Positioner

XPLORE 1.2™ PQMS system comes with an automatic positioner which moves the susceptibility insert in order to align the sample with either of the secondary coils. The positioner is driven by a stepper motor which is controlled from an electronic control unit. The sample positioner module consist of the stepper motor, the positioner assembly, the electronic control unit, and its command utility.

Although presently the sample positioner is customized for susceptibility measurements, one can do slight modifications to use the assembly for other experimental purposes as well.

The hardware is shown in Figure 4.1. The screen shot of the associated command utility is shown in Figure 4.2.

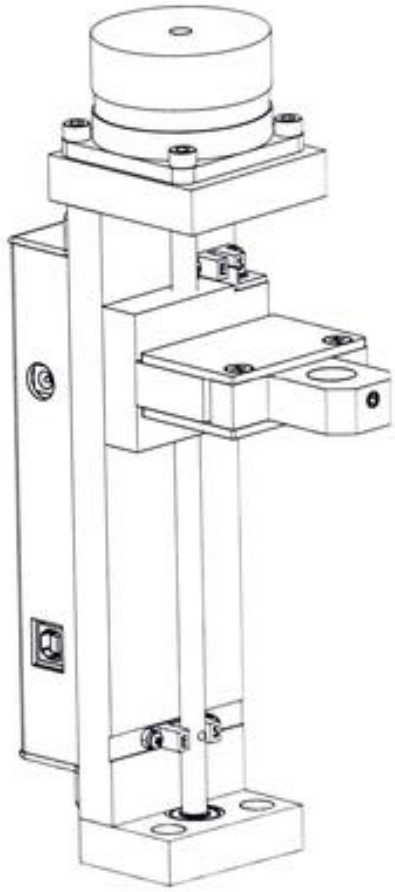


Figure 4.1: Sample positioner unit

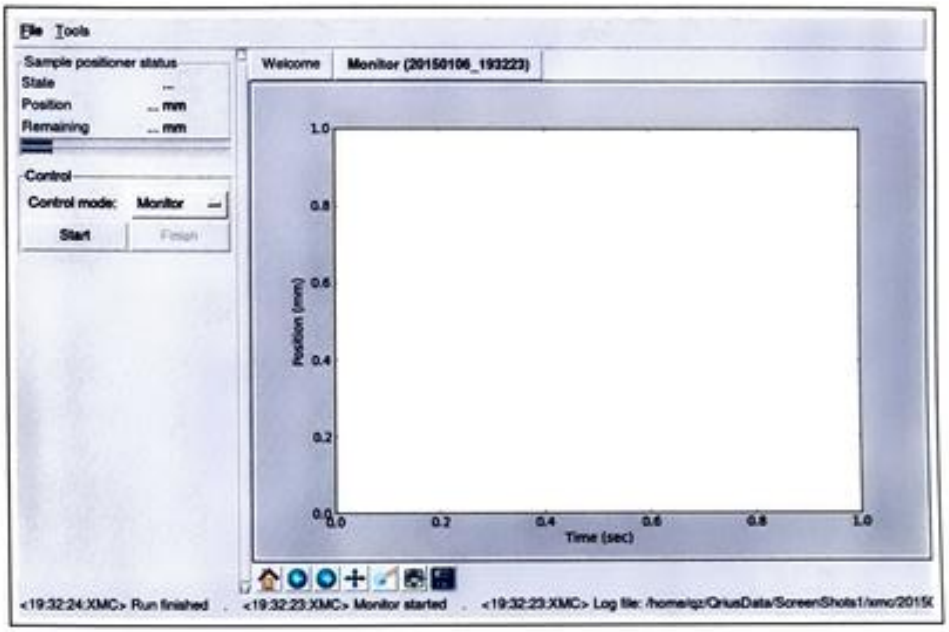


Figure 4.2: A screen shot of the command utility for the sample positioner. Various commands to move the sample positioner can be accessed by opening their respective settings dialog box from the **Settings** menu on the menu bar.

4.1 Movement commands

The command utility of the sample positioner provides 4 commands related to movement:

- **Reset**, which moves the sample to the topmost position, which is also the absolute zero. All the distances are measured are relative to this position in downward direction. To reset the device click **Tools → Reset**.

- **Move absolute**, which moves the sample to a specified distance from the absolute zero in the downward direction. To use it open the **Move absolute** dialog box (Figure 4.3a) by clicking **Tools → Move absolute**. Then specify the position at which the sample has to be placed in the **Position** field. Then click **File → Apply** to confirm the position, or **File → Cancel** to cancel it.

- **Move relative**, which moves the sample to a specified distance relative to its current position. To use it open the **Move relative** dialog box (Figure 4.3b) by clicking **Tools → Move relative**. Then specify distance by which the sample has to be moved in **Distance** field. Note that a positive distance will move the sample in *downward* direction, while negative distance will move it upwards. Finally click **File → Apply** to confirm the position, or **File → Cancel** to cancel it.

- **Stop**. To stop the the sample positioner if it is moving click **Tools → Stop**.



(a) (b)

Figure 4.3: Dialog boxes to move the sample **(a)** relative to the zeroth position and **(b)** relative to the current position.

4.2 Data files

As soon as a run is initiated, a log file is opened in the XLIA's default data folder, **/home/pqms/QriusData/Sample1/xmc**. The name of the file is derived from current date **YYYYMMDD** and time **HHMMSS** as **YYYYMMDD_HHMMSS_Mode.csv**, where **Mode** can be either **V_ Time** or **VF**.

i A few lines at the beginning of the file contains information about sample details and column headings. These lines start with a #. Thereafter each row contains time stamp and sample position separated by comma.

i The default destination path of the recorded data can be changed by an user using the following guidelines.

- **/home/pqms** is the home folder corresponding to **pqms** login id. If some other login id is used, the path changes accordingly.
- **QriusData** is the root of the default destination folder. This can be changed from **Settings → Global settings** in the menu bar of Qrius master window.
- **Sample1** is the default sample folder generated by concatenation of default sample name (**Sample**) and sample id (**1**). This can be changed from **Sample settings** dialog from the Qrius master window.
- **xmc** represents a folder dedicated for sample positioner unit. **It cannot be changed.**

i The recorded file, being in **csv** format, can be directly opened in OpenOffice, Excel, Origin, Matlab, and other popular analysis tools. At the end of a run, another file with **xpl** extension is created, which can be opened by Qrius Analyzer.

4.3 Sample positioner specifications

Parameter	Value
Stoke length	60 mm
Positioning accuracy	Better than 100 micron
Speed	2 mm/sec
Stall detection	Yes
Computer connectivity	USB

System Preparation

5.1 Parts identification

Before preparing the system for cooling please the user should be familiar with the different components, and their function, of the cryostat and the exchange gas flushing system.

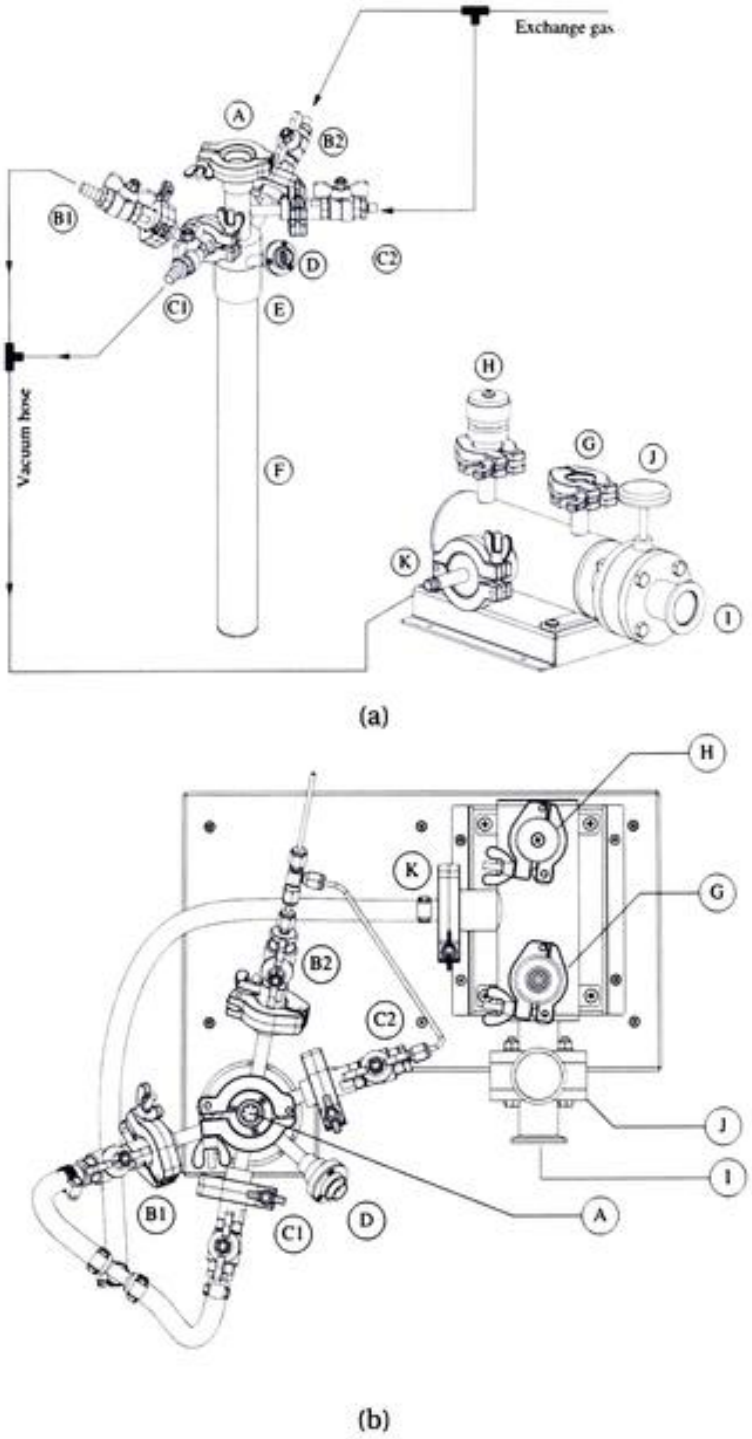


Figure 5.1: Cryostat and vacuum manifold. **(a)** Side view, **(b)** Top view

Item	Description
A	Sample insertion port.
B1	Heater chamber evacuation port.
B2	Heater chamber flushing port.
C1	Sample chamber evacuation port.
C2	Sample chamber flushing port.
D	Heater electrical connection feed-through.
E	Cryostat stand.
F	Non-magnetic double-walled cryostat.
G	Pirani gauge mounting port.
H	Release valve.
I	Vacuum pump connection port.
J	Butterfly valve.
K	Cryostat vacuum hose connection port.

Table 5.1: Different components of the cryostat and exchange gas flushing system

5.2 Flushing with exchange gas

Exchange gas is used to create a good thermal contact either between the sample and the heater or the heater and the environment. However, if thermal insulation is required in a particular chamber, then that chamber is completely evacuated.

Before flushing the system make sure that the sample insert is in place and all the clamps with their o-rings inside are tightly closed.



To **close** any of the valves mentioned in below, turn it **clockwise**. To open, turn it anti-clockwise.

Procedure:

1. Close all the valves (**B1**, **B2**, **C1**, **C2**, **H** and **J**) and power on the vacuum pump.
2. Open the butterfly valve **J**. Make sure that the air pressure is falling on the Pirani gauge.
3. Open the evacuation port of the chamber which needs to be evacuated/flushed, **B1** for the heater chamber and **C1** for the sample chamber. If both need to be evacuated/flushed, then open both the evacuation ports. The pressure on the Pirani gauge will rise the moment the ports are opened. Wait for the pressure to fall again.
4. If there is any chamber which needs to be kept evacuated, then close its evacuation port. Thereafter, close the butterfly valve **J** to cut-off the pump.

5. Make sure that the inlet pressure of the exchange gas is less that 10 psi. Then, open the flushing port(s) **B2** or **C2**, of the chamber(s) that needs to be flushed. After a few seconds close the flushing port(s).

6. Remove the excess pressure by opening the release valve **H**. Close the valve immediately.

7. Close both evacuation ports, if open, and open the butterfly valve **J**.

8. Finally, if there is any chamber that need to be kept evacuated then open its evacuation port. Do not turn off the pump or close the butterfly valve **J**.

By following the above procedure the system is ready to be inserted in the liquid nitrogen dewar.

⚠ Prior to opening any of the inlet ports **B2** or **C2**, make sure that both the release valve **H** and the butterfly valve **J** are closed. Otherwise a lot of exchange gas will be escape to the atmosphere.

5.3 Cooling with liquid nitrogen

Using liquid nitrogen and exchange one can achieve temperature range of 77 kelvin (-196°C) to 450 (+177°C) kelvin. The only thing to note is that for temperatures below 200K (-73°C) a good thermal contact is needed between the liquid nitrogen and the sample. Therefore exchange gas should be present in both the sample and heater chambers. For temperatures above 200K the heater chamber needs to be evacuated so that all the heat goes to the sample only. The sample chamber still needs to have exchange case.

Temperature Range	Heater Chamber	Sample Chamber
77K - 200K	Helium	Helium
200K - 450K	Vacuum	Helium

Table 5.2: Exchange gas combinations

i There is no need to have separate acquisition runs for temperature less than 200K and for temperature greater than 200K. The heater chamber can be evacuated during the experiment by simply opening the evacuation port **B1** (assuming that vacuum pump is running and the butterfly valve **J** is open).

⚠ Do not cool the cryostat if air is present in any of the chambers. The moisture present in the air will condense and may cause deterioration inside.

5.4 System shutdown

If the system has to be shutdown or sample needs to be removed then follow the following procedure:

1. Take out the cryostat from the dewar and place it on the manifold base.
2. Close the butterfly valve **J** to cut-off the pump.
3. Open both the evacuation ports. Make sure that there is some exchange gas in both the chambers.
4. Let the system heat up to room temperature. Sample heater can be used to quicken the process.
5. Open the release valve to let air in.
6. The sample insert can now be safely removed.
7. If the system is not being used further put one of sample inserts back into the cryostat. Do not leave the insertion port open.
8. Close all the valves and switch off all the power supplies.

Electrical Transport Property Measurement

6.1 Sample insert

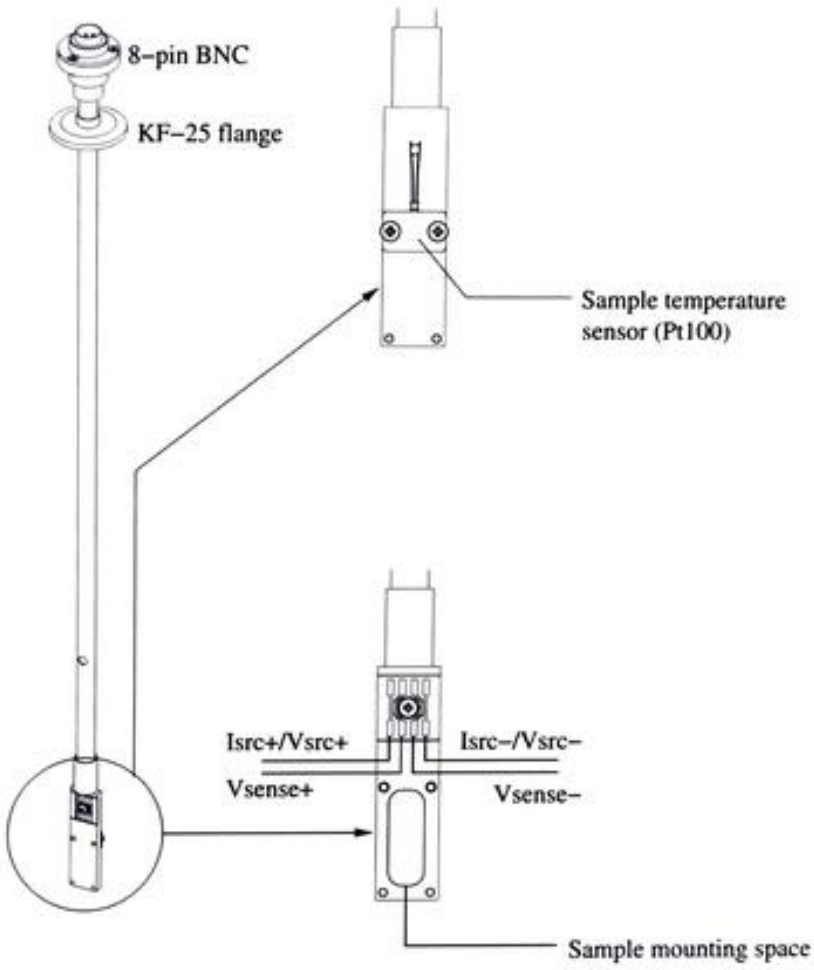


Figure 6.1: I-V and R-T measurement insert

6.2 Mounting sample

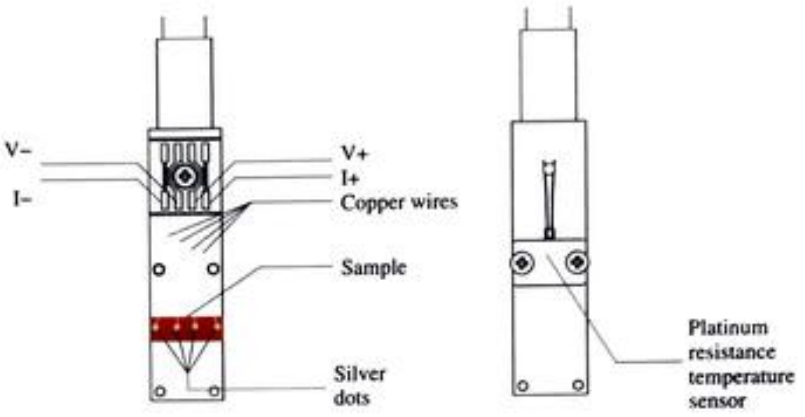


Figure 6.2: Sample mount for I-V and R-T measurement

- Paste the sample on the sample mount as shown in Figure 6.2. Position it as close to the thermocouple as possible to avoid lag in temperature readout.
- Solder four thin wires to the contact pads I+, I-, V+, and V-.
- Connect their other ends to the sample using silver epoxy, indium solder or any other technique, which ensures a good electrical contact of the wires to the sample.
- Gently slide down the four-probe measurement insert in the cryostat and clamp it using its top flange (KF-25).
- Connect the insert with the XSMU and XTCON as shown in Figure 6.3.

6.3 Electrical connection

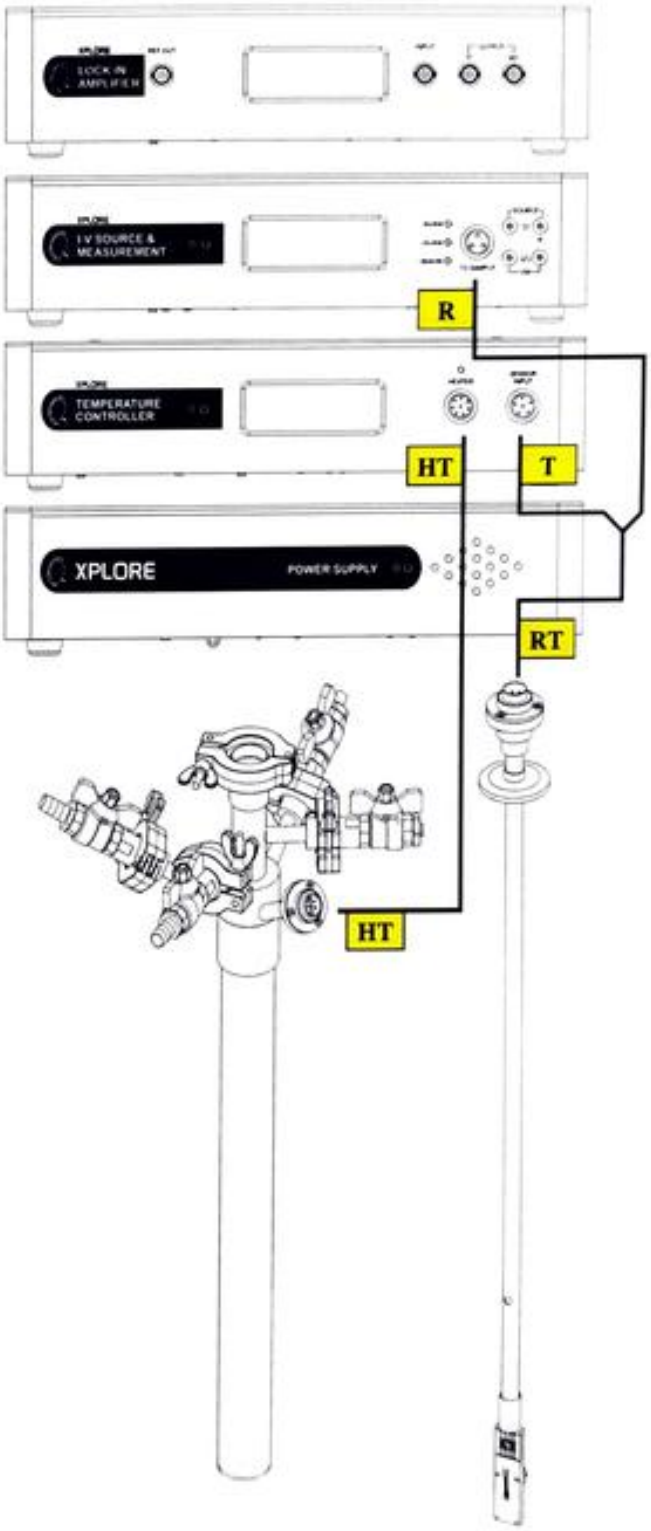


Figure 6.3: Cable connection for I-V and R-T experiments

6.4 Acquiring I-V data



Before taking an I-V plot, log I-V data to ensure that the contacts are fine.

- Launch Qrius and click **Measurement Mode Settings**. Open **Electrical DC conductivity (I-V & R-T)** module as shown in Figure 6.4.
- Click **Settings → I-V Source and measurement unit → I-V ramp settings** to open the I-V ramp settings dialog box. Fill the details of the I-V ramp as described in Section 2.3.
- Click **Settings → Temperature Controller → Stepped ramp settings** to open the stepped ramp settings dialog box. Fill the details of the stepped ramp as described in Section 1.4.
- In the module window select **I-V (stepped ramp)** as **Run Mode**.
- Start I-V acquisition run by pressing the **Start** button. If the acquisition run has to be terminated manually then press **Finish**.
- I-V acquisition run will start and the data will be plotted alongside.
- When finished a message will appear on the status bar mentioning the name and location of the saved data file.

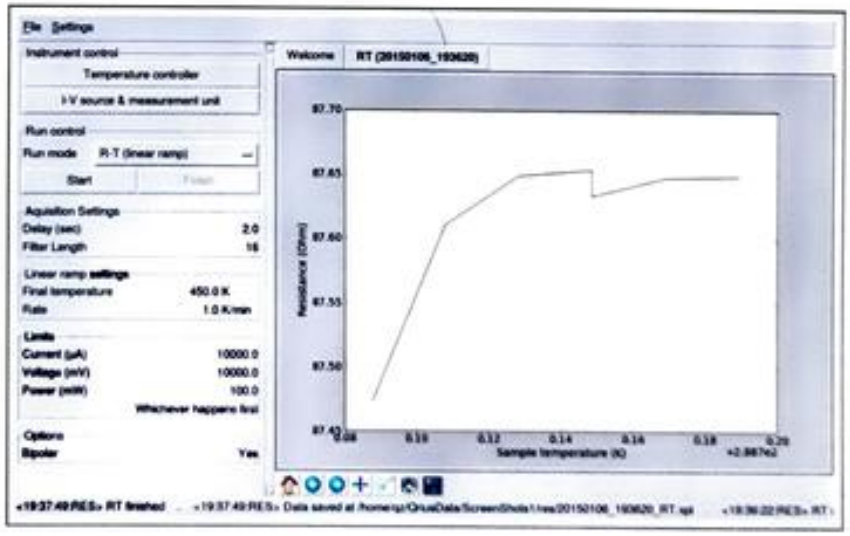


Figure 6.4: I-V data acquisition window

6.5 Acquiring R-T data



Before taking an R-T plot, log I-V data to ensure that the contacts are fine.

- Launch Qrius and click **Measurement Mode Settings**. Open **Electrical DC conductivity (I-V & R-T)** module as shown in Figure 6.4.

- Click **Settings → I-V Source and measurement unit → Resistance measurement settings** to open the resistance measurement settings dialog box. Fill the details of the resistance measurement as described in Section 2.2.

- If temperature has to be changed linearly, click **Settings → Temperature Controller → Linear ramp settings** to open the linear ramp settings dialog box. Fill the details of the linear ramp as described in Section 1.3.

  If temperature has to be increased in steps, click **Settings → Temperature Controller → Stepped ramp settings** to open the stepped ramp settings dialog box. Fill the details of the stepped ramp as described in Section 1.4.

- In the module window select **R-T (linear ramp)** or **R-T (stepped ramp)** as **Run Mode**.

- Start *R-T* acquisition run by pressing the **Start** button. If the acquisition run has to be terminated manually then press **Finish**.

- *R-T* acquisition run will start and the data will be plotted alongside.

- When finished a message will appear on the status bar mentioning the name and location of the saved data file.

6.6 Data files

As soon as a run is initiated, 3 log files are opened. One the XSMU's default data folder (Section 2.8), second in the temperature controller data folder (Section 1.7), and the third in the default data folder for electronic transport property data folder (**/home/pqms/QriusData/Sample1/res**). The name of the file is derived from current date **YYYYMMDD** and time **HHMMSS** as **YYYYMMDD_HHMMSS_Mode.csv**, where **Mode** can be one of the either **RT** or **IV_T**.

The files created in the **res** folder contains data files which are created by amalgamating the relevant data from the XSMU data (Section 2.8) and temperature controller data (Section 1.7). In the amalgamated data, each row contains current, measurement voltage, resistance, sample temperature, and the heater temperature separated by comma.



The recorded file, being in **csv** format, can be directly opened in OpenOffice, Excel, Origin, Matlab, and other popular analysis tools. At the end of a run, another file with **xpl** extension is created, which can be opened by Qrius Analyzer.

6.7 Save and recall settings

It is always useful to save instrument settings for future reference. This is done through method files. A method file is identified with its **xmt** extension. To create a method file, first set all parameters using different dialog boxes described in previous sections, and click **File → Save method** in the menu bar of the XSMU utility window (Figure 2.2). Enter the file name in the opened dialog. To load a stored method, use **File → Open method**.



By default the method file goes to **/home/pqms/QriusData/Sample1/method** folder.
This can be changed in a similar way described for the data files.



If the method file is not for dc measurements, an error message "**Method failed. Not a resistivity method**" is shown in the status bar.

Magnetic AC Susceptibility Measurements

7.1 Sample insert and coil assembly

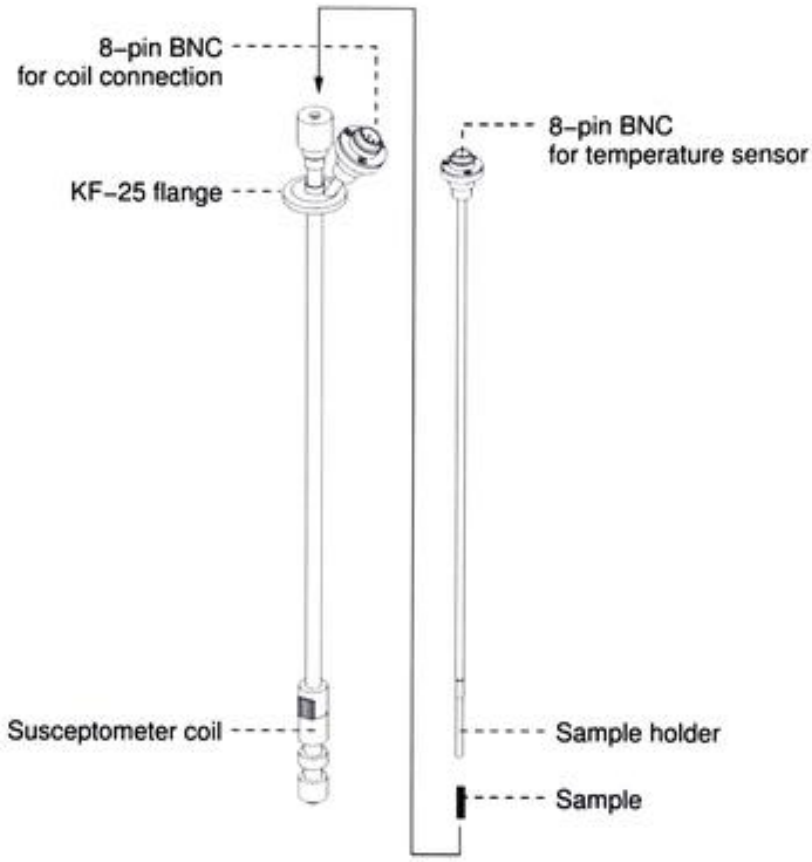


Figure 7.1: AC susceptibility measurement insert

7.2 Mounting Sample

- Lower the coil insert into the cryostat without the sample insert.
- Encapsulate the sample in a butter-paper wrapping and insert it into the sample holder pipe sleeve.
- Push the sample in the sleeve until it just touches the sample temperature sensor.
- See to it that no other magnetic material is introduced in the coil except the test sample.
- Lower the sample insert into the coil insert and tighten the nut holding the sample insert.
- Check all the cabling. Make sure that the coil insert is connected to the lock-in amplifier and the sample insert and the heater coil is connected to the temperature controller.

7.3 Electrical connection

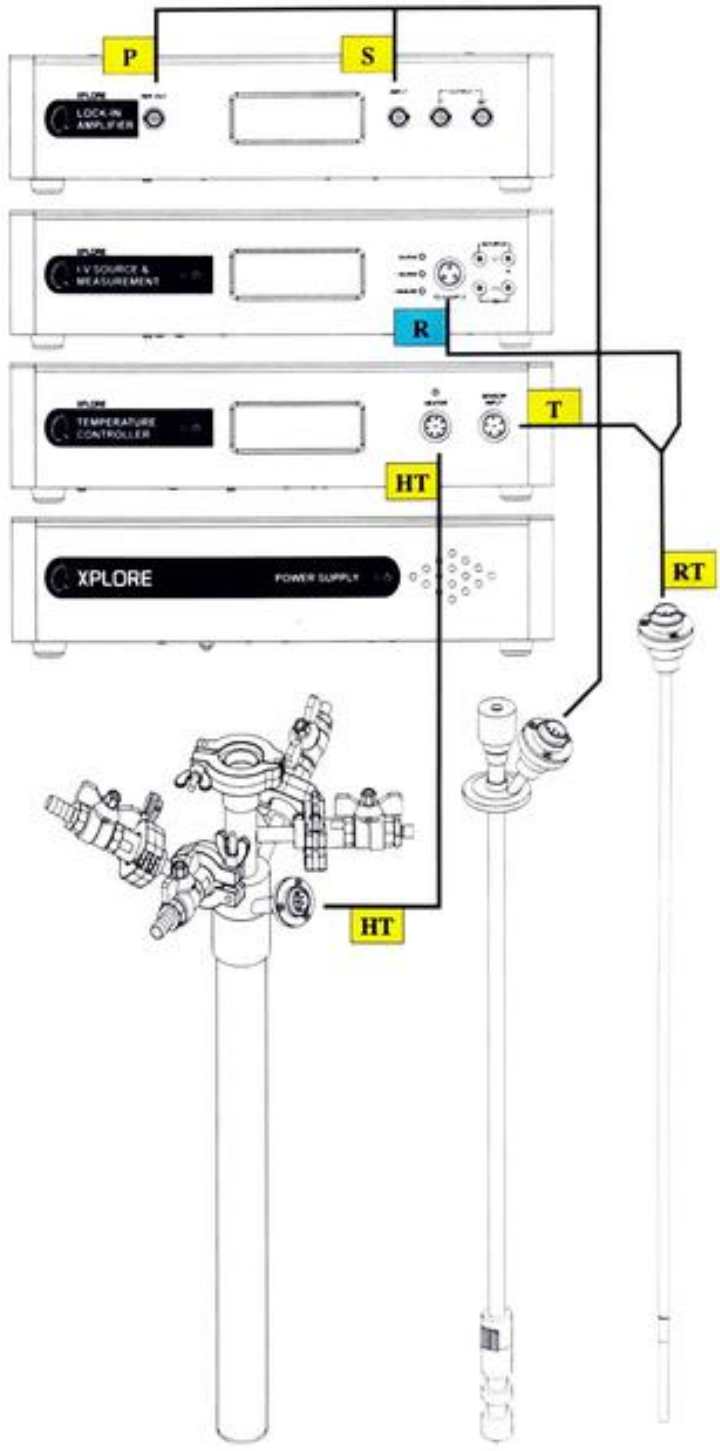


Figure 7.2: Electrical connection for AC susceptibility measurement.

7.4 Sample positioning

Open the susceptibility measurement module by clicking **Measurement Mode Settings → Magnetic AC Susceptibility**. In the module window select χ - **L** as **Run mode** and select start. This acquisition run will measure and plot the value of χ at different positions. From this plot, the two position where the secondary coil pick up is strongest will be calculated and remembered.

Once the two positions are known, the value of susceptibility used thereafter will be the difference of the susceptibilities measured at the those two positions.

If the two positions of maximum secondary coil signal are known previously, then the values can be manually entered. To do that click **Settings → χL acquisition setting** and the the two position in the **Probe up (mm)** and **Probe down (mm)** fields respectively (Figure 7.3).

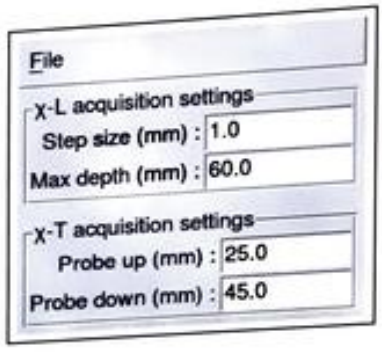


Figure 7.3: Dialog box for setting sample position

7.5 Acquiring χ-T data

- Prepare the system as described in Section 5.2.
- Bring the temperature to a point where sample is expected to show some amount of AC susceptibility.
- Perform a χL run (Section 7.4) to know the positions of maximum secondary coil signal.
- Click **Settings → Lock-in Amplifier → Reference settings** and enter the settings for reference generator (Section 3.3). The value of **Amplitude** in the above will be overridden with the one in the next step.



- Click **Settings → Lock-in Amplifier → Acquisition settings** and enter the settings for the drive mode (Section 3.6). It is highly recommended to use current drive in the above step since amplitude of magnetic field is dependent current and not voltage.



- If temperature has to be changed linearly, click **Settings → Temperature Controller → Linear ramp settings** to open the linear ramp settings dialog box. Fill the details of the linear ramp as described in Section 1.3.

If temperature has to be increased in steps, click **Settings → Temperature Controller → Stepped ramp settings** to open the stepped ramp settings dialog box. Fill the details of the stepped ramp as described in Section 1.4.

- In the module window select χ-**T (linear ramp)** or χ-**T (stepped ramp)** as **Run Mode**.
- Start χ-T acquisition run by pressing the **Start** button. If the acquisition run has to be terminated manually then press **Finish**.
- χ-T acquisition run will start and the data will be plotted alongside.
- When finished a message will appear on the status bar mentioning the name and location of the saved data file.

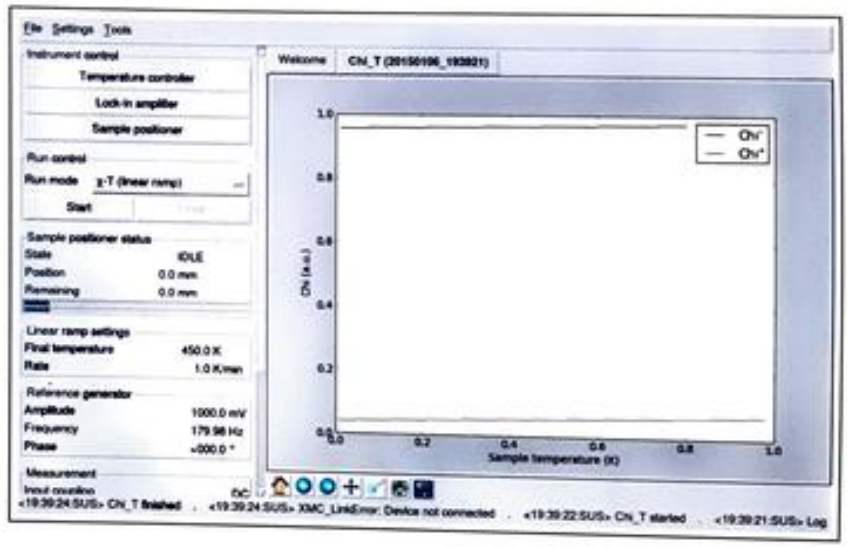


Figure 7.4: Susceptibility measurement module

7.6 Theory of operation

In AC susceptibility measurement apparatus we have a primary coil around which are two sets secondary coil, one of which is wounded clockwise and the other anticlockwise. When primary coil is excited with an AC current, an emf is generated in the secondary coils due to electromagnetic induction. When no sample is kept then the emf generated in the secondary coils are equal and opposite. When a magnetic sample is kept in one of the secondary coils, it changes the flux passing through that coil. Hence, a net emf is created which is measured.

The current given to the primary coil is of the form:

$$I(t) = I_0 \sin(\omega t) \tag{7.1}$$

This generates a magnetic field ($B(t)$) in the primary coil:

$$\begin{aligned} B(t) &= \mu_0 n_p I_0 \sin(\omega t) \\ &\propto \mu_0 I_0 \sin(\omega t) \end{aligned} \tag{7.2}$$

where n_p is the number of turns in primary coil.

Total calculate total flux through one of the secondary coils we will have to integrate the field $B(t)$ over the cross-sectional area of the coil. We can proceed by assuming the the $B(t)$ is constant in space. :

$$\begin{aligned} \Phi_1(t) &= \int B(t)dA \\ &\propto B(t)\int dA \\ &\propto B(t)\mu_0 I_0 \sin(\omega t) \end{aligned} \tag{7.3}$$

here, the proportionality constant will depend on the cross-sectional areas of primary and secondary coils.

Similarly, the total flux through the other secondary coil Φ_2 is equal to $-\Phi_1$.

Now we calculate the emf generated due to the flux in one of the secondary coils.:

$$\begin{aligned} \text{emf}_1(t) &\propto \mu_0 n_s \frac{d\Phi_1(t)}{dt} \\ &\propto \mu_0 I_0 \frac{d\sin(\omega t)}{dt} \\ &\propto -\mu_0 I_0 \omega \cos(\omega t) \end{aligned} \tag{7.4}$$

where n_s is the number of turns of the secondary coil.

Like wise, the emf generated in the other secondary coil (emf_2) is equal to $-\text{emf}_1$.

Now, if a sample of susceptibility χ is kept in one of the secondary coils then the permittivity in that coil changes to $\mu = \mu_0(1+\chi)$.

Therefore total emf generated in the secondary coil is:

$$\begin{aligned} \text{emf}(t) &= \text{emf}_1(t) + \text{emf}_2(t) \\ &\propto -\mu\, I_0 \omega \cos(\omega t) + \mu_0\, I_0 \omega \cos(\omega t) \\ &\propto -\mu_0(1+\chi)\, I_0 \omega \cos(\omega t) + \mu_0\, I_0 \omega \cos(\omega t) \\ &\propto -\mu_0 \chi\, I_0 \omega \cos(\omega t) \\ &= -C\mu_0 \chi\, I_0 \omega \cos(\omega t) \end{aligned} \tag{7.5}$$

where, proportionality constant C is called geometric factor.

It is this emf which is measured at the ends of the secondary coil.

In all magnetic materials there is some amount of magnetic hysteresis. Therefore, when one applies a periodically changing magnetic field

there is some amount of energy loss. In AC susceptibility measurements this causes a certain amount of phase shift between primary coil current I and secondary coil emf emf.

This phase shifted is accounted in theory by assuming the susceptibility χ to have a non real component. Thus:

$$\begin{aligned} \chi &= \chi' + i\chi'' \\ &= |\chi|\cos(\theta) + i|\chi|sin(\theta) \\ &= |\chi|e^{i\theta} \\ \text{where } |\chi|^2 &= \chi'^2 + \chi''^2 \end{aligned} \tag{7.6}$$

Now we assume an complex emf whose real part is the one which is measurable. Thus the equation becomes:

$$\begin{aligned} \text{Re (emf}(t)) &= -C'\mu_0\chi\, I_0\omega\cos(\omega t) \\ &= -CI_0\omega\,\text{Re}\,(\chi e^{i\omega t}) \\ &= -CI_0\omega\,|\chi|\text{Re}\,(e^{i\theta}e^{i\omega t}) \quad \text{using } \chi = |\chi|e^{i\theta} \\ \Rightarrow \text{emf}(t) &= -CI_0\omega\,|\chi|e^{i(\omega t+\theta)} \\ \Rightarrow \text{Re (emf}(t)) &= -CI_0\omega\,|\chi|\cos(\omega t+\theta) \end{aligned} \tag{7.7}$$

Expanding the cos term we get:

$$\text{Re (emf}(t)) = -CI_0\omega\,|\chi|\,(\cos\omega t\cos\theta - \sin\omega t\sin\theta) \tag{7.8}$$

This θ comes out as the phase difference between the primary current and secondary emf.

To measure AC susceptibility, we pass the secondary coil signal to the input of of the lock-in amplifier. Since we already know that the time period of the input signal is same as that of the reference signal, we can use Fourier expansion to expand the input signal $\mathbf{U_{in}}$ around the reference frequency ω. :

$$\begin{aligned} \mathbf{U_{in}}(t) &= a_1\cos(\omega t) + a_2\cos(2\omega t) + ... + a_n\cos(n\omega t) + \\ & \quad b_1\sin(\omega t) + b_2\sin(2\omega t) + ... + b_n\sin(n\omega t) \end{aligned} \tag{7.9}$$

In the lock-in amplifier the in-phase measurement channel measures the a_1 term in the equation 7.9, and quadrature measurement channel measures the b_1 term. Both the values are in the form of a DC voltage. The lock-in is able to filter out any contribution due to any frequency other than ω.

Comparing with equation 7.8, we can find the value of susceptibility:

$$
\begin{aligned}
a_1 &= -CI_0\omega|\chi|\cos(\theta) & b_1 &= CI_0\omega|\chi|\sin(\theta) \\
&= -CI_0\omega\chi' & &= CI_0\omega\chi'' \\
\Rightarrow C\chi' &= -\tfrac{a_1}{I_0\omega} & \Rightarrow C\chi'' &= \tfrac{b_1}{I_0\omega}
\end{aligned}
$$

Note that the geometric factor C is dependent on many terms and is difficult to accurately calculate. But since it is a constant for a particular system it has no effect on the qualitative nature of susceptibility. For different systems, C is expected to be same within an order of magnitude.

System Specification

A.1 Cryostat and temperature controller

Parameter	Value
Sample chamber	
Dimension	21 mm (dia) × 100 mm (length), top loading
Construction	Non-magnetic, double walled
Temperature range	
Temperature range	80 K – 450 K
Cryogen	Liquid N_2
Exchange gas	Air, port available for evacuation/flushing
Temperature measurement and control	
Sensors	Pt100 and K-type thermocouple
Temperature resolution	0.01 K
Temperature stability	Better than ±0.1 K in isothermal mode
Temperature ramp-rate	1 K/min – 10 K/min in steps of 0.1 K/min
Control algorithm	PID
PID parameters	User configurable
Heater	
Heater power	40 Watt
Winding type	Non-magnetic
Others	
Computer connectivity	USB

A.2 I-V source and measurement unit for 4-probe R-T

Parameter	Value
Source mode	Current or voltage
Current source specification	
Current source range(s)	100 μA, 1 mA, 10 mA
Current set-point resolution	Better than 0.05 % of full-scale
Voltage compliance	10V

I-V source and measurement unit specification continued on next page ...

... I-V source and measurement unit specification continued from previous page

Parameter	Value
Voltage source specification	
Voltage source range(s)	10 V
Voltage set-point resolution	Better than 0.005 % of full-scale
Current compliance	10 mA
Ammeter specification	
Current measurement range(s)	100 μA, 1 mA, 10 mA
Current measurement resolution	6½ digit
Voltmeter specification	
Voltage measurement range(s)	100 mV, 1 V, 10 V
Voltage measurement resolution	6½ digit
Input impedance	$> 10^{12}\,\Omega$
Ohmmeter specification	
Resistance measurement range(s)	10 Ω, 100 Ω, 1 kΩ, 10 kΩ, 100 kΩ, 1 MΩ, 10 MΩ
Resistance measurement resolution	6½ digit
Others	
Computer connectivity	USB

A.3 Lock-in amplifier for AC-susceptibility (χ-T) measurement

Parameter	Value
Sample	
Sample size	5 mm (dia) × 10 mm (length)
Field strength	4 Oe RMS field at 80 Hz
Primary driver	
Frequency range(s)	10 Hz – 10 KHz
Reference output	2V peak-to-peak
Reference generation	Direct digital synthesis
Lock-in detection	
Integration time	1 sec
Pre-amp gain(s)	1, 10, 100
Post-amp gain(s)	1, 10, 100
Measurement	

AC-susceptometer specification continued on next page ...

... AC-susceptometer specification continued from previous page

Parameter	Value
Measurement resolution	16-bit
Measured quantities	Both amplitude and phase
Others	
Computer connectivity	USB

A.4 Data acquisition and control software: Qrius 1.1

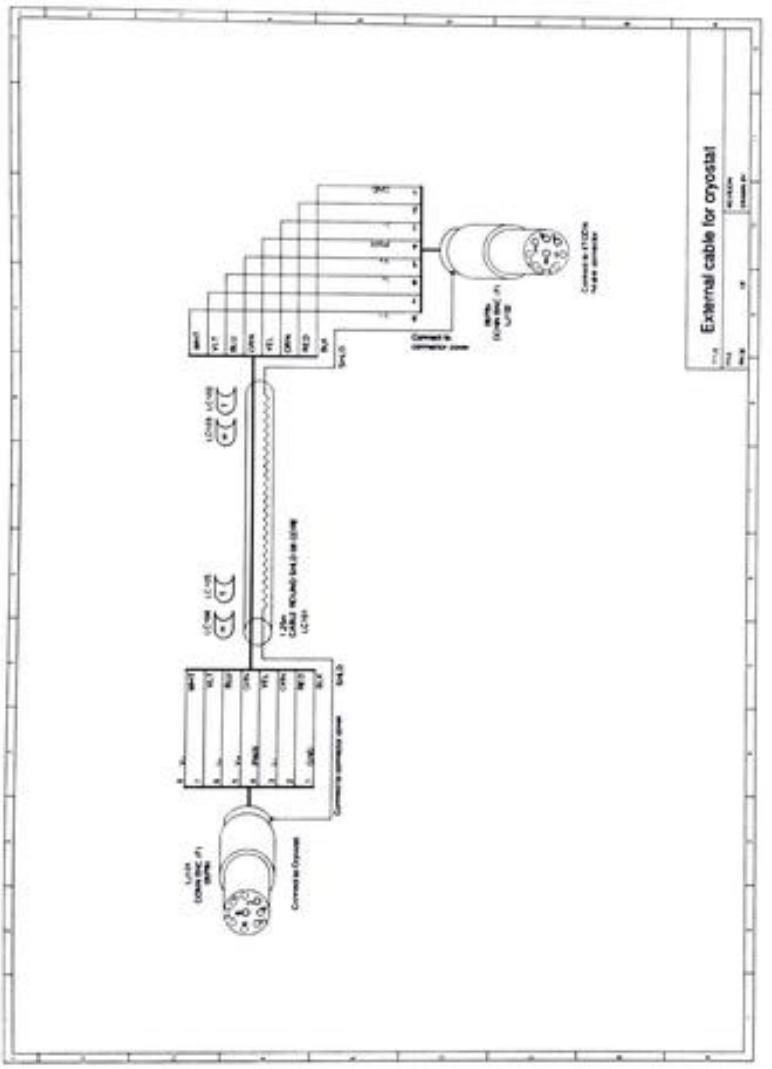
Computer automation of R-T and χ-T measurements using above mentioned modules have been provided by Qrius 1.1 software. It allows easy control of all experimental parameters and real time recording and plot of physical quantities. The software runs on Ubuntu 12.04.

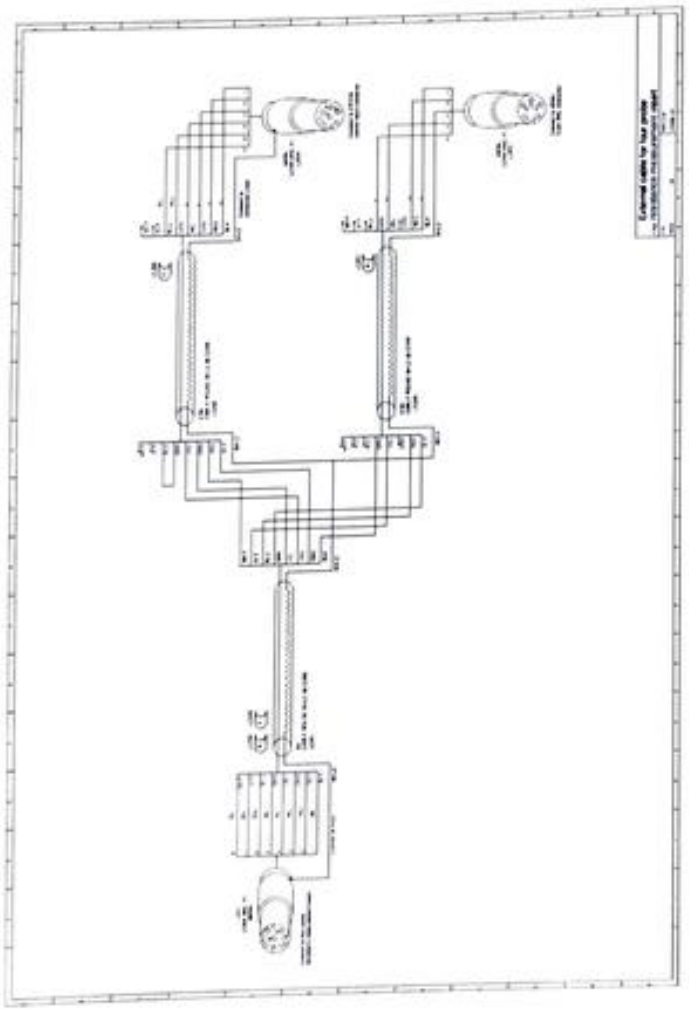
A.5 Other user provided accessories required to run the system

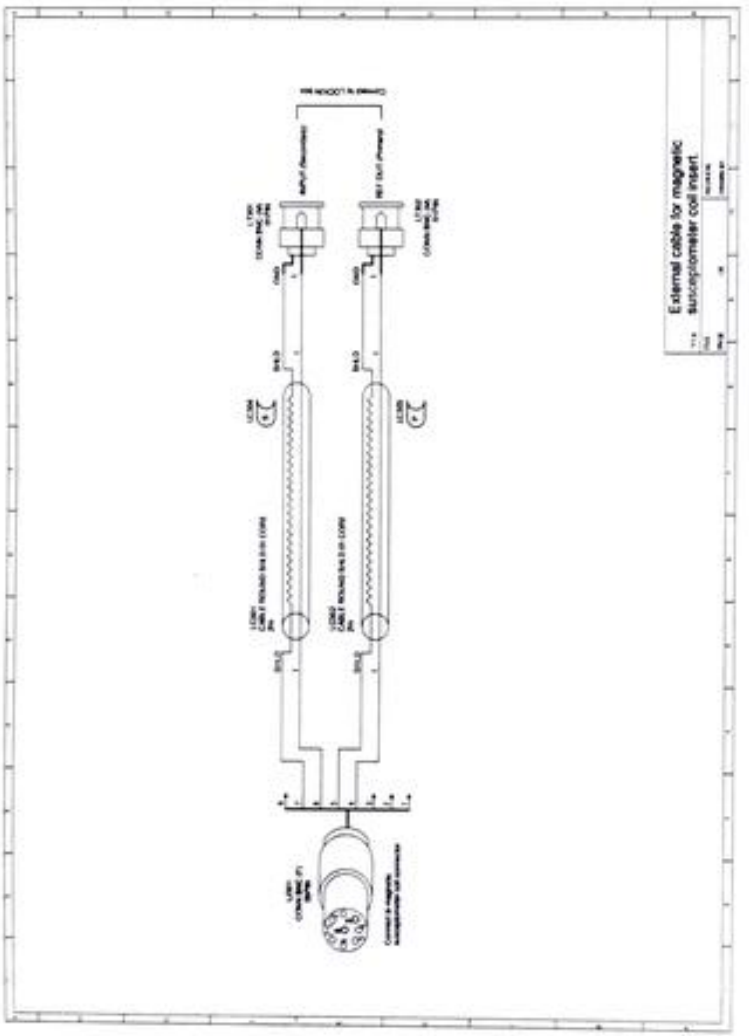
The following is a list of accessories that *need to be provided by the user* for proper operation of XPLORE-1.1 PQMS setup.

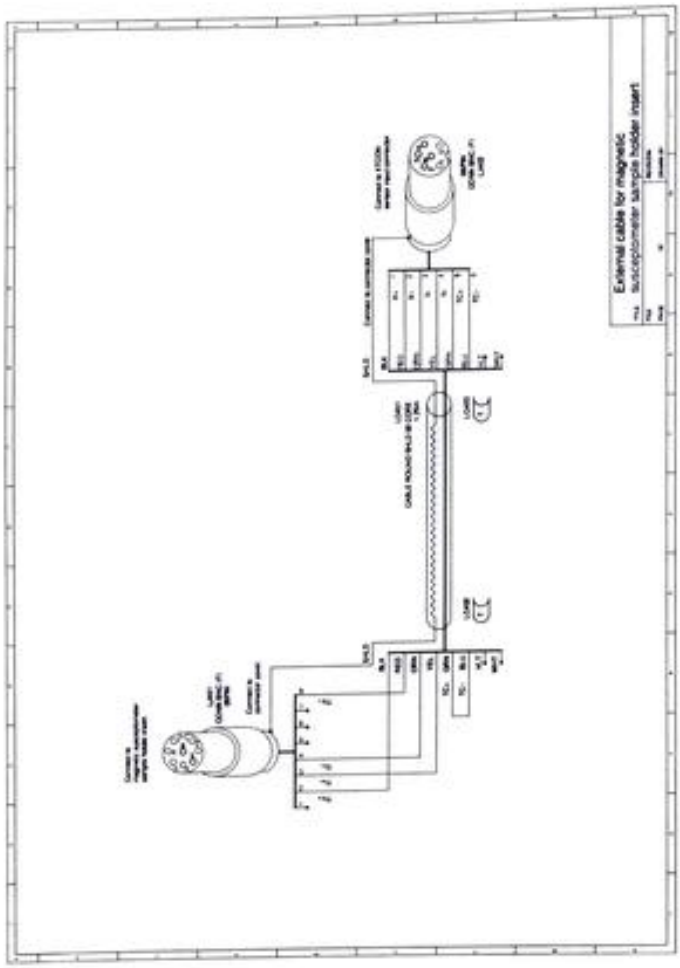
- Rotary vacuum pump.
- Minimum of 3 liters of liquid nitrogen.
- Computer with DVD drive, USB 2.0 port, 2GB RAM, 100GB of hard-disk space and Pentium dual core or above processor.

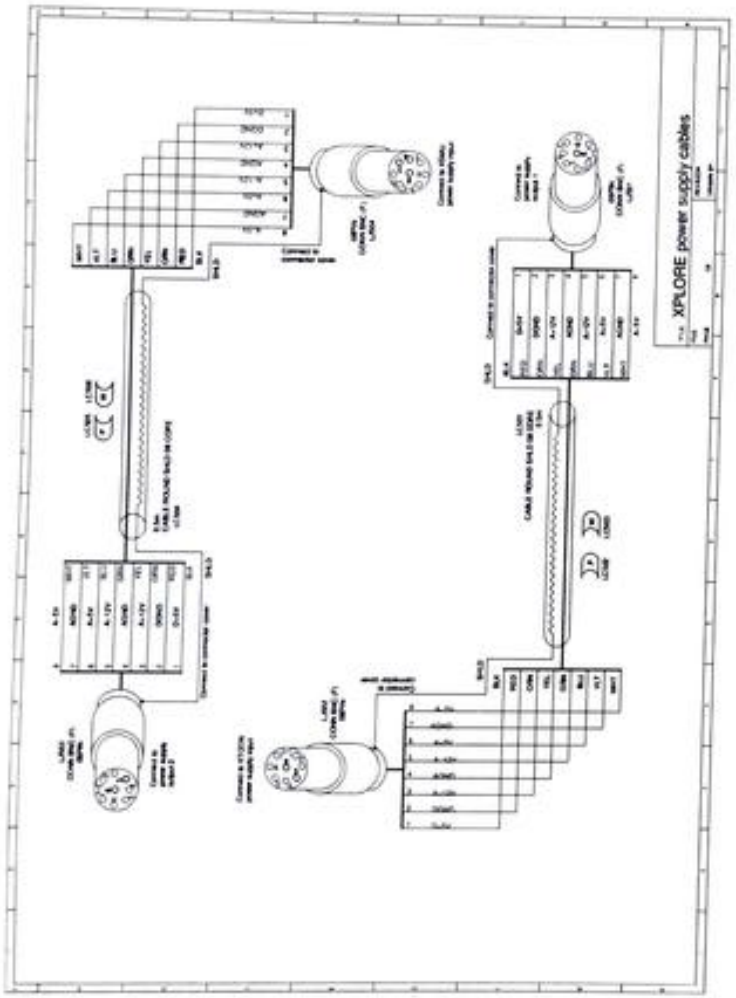
Cable Details


External cable for cryostat




External cable for magnetic
susceptometer coil insert


External cable for magnetic
susceptometer sample holder insert


XPLORE power supply cables